CS 498: Machine Learning System Spring 2025

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The Grainger College of Engineering

Today

DL Inference

- FlashAttention
- LLM Inference
- Continuous Batching

FlashAttention

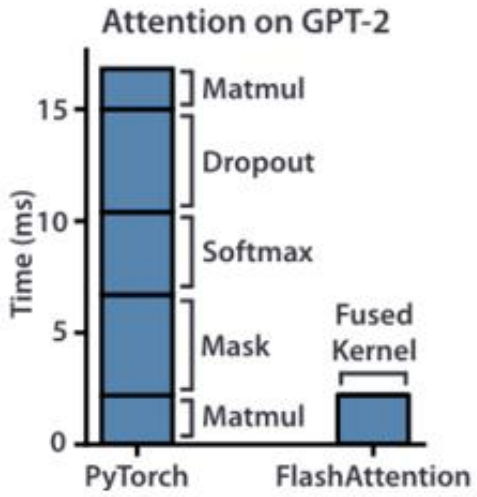
- First introduced at HAET workshop @ICML July 2022
- Published @ NeurIPS Dec 2022
- Very useful even though many people probably don’t even know they are using it!

FlashAttention: Fast and Memory-Efficient Exact Attention with IO-Awareness

Tri Dao, Dan Fu ({trid, danfu}@cs.stanford.edu)
7/23/22 HAET Workshop @ ICML 2022

Tri Dao, Daniel Y. Fu, Stefano Ermon, Atri Ruda, Christopher Ré. Flash Attention: Fast and Memory-Efficient Exact Attention with IO-Awareness. *arXiv preprint arXiv:2205.14135.* https://github.com/HazyResearch/flash-attention.

FlashAttention



FlashAttention

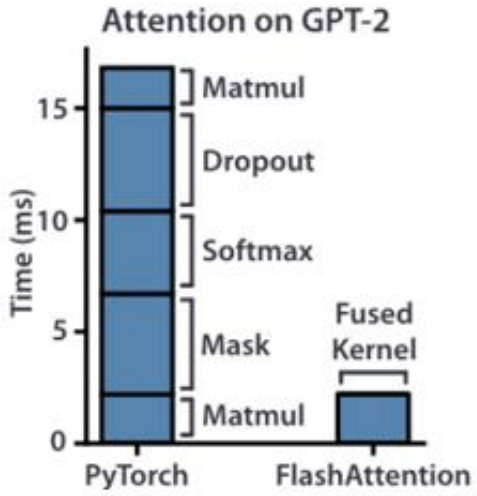


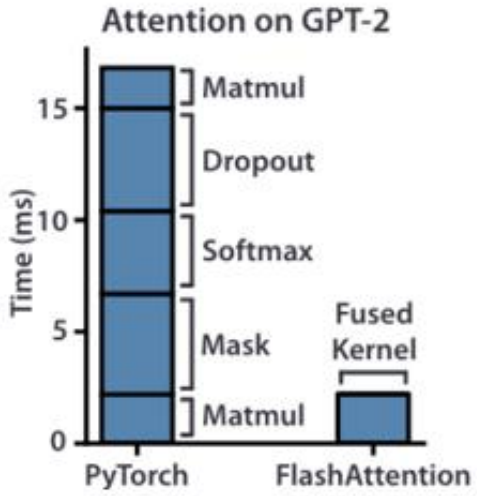
Table 1. Proportions for operator classes in PyTorch.

Operator class	% flop	% Runtime
△ Tensor contraction	99.80	61.0
□ Stat. normalization	0.17	25.5
○ Element-wise	0.03	13.5

- Matrix multiplication takes up 99% of the FLOPS
- But only takes up 61% of the runtime

Question: Why do other operators take so much time?

FlashAttention



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Question: Why do other operators take so much time?

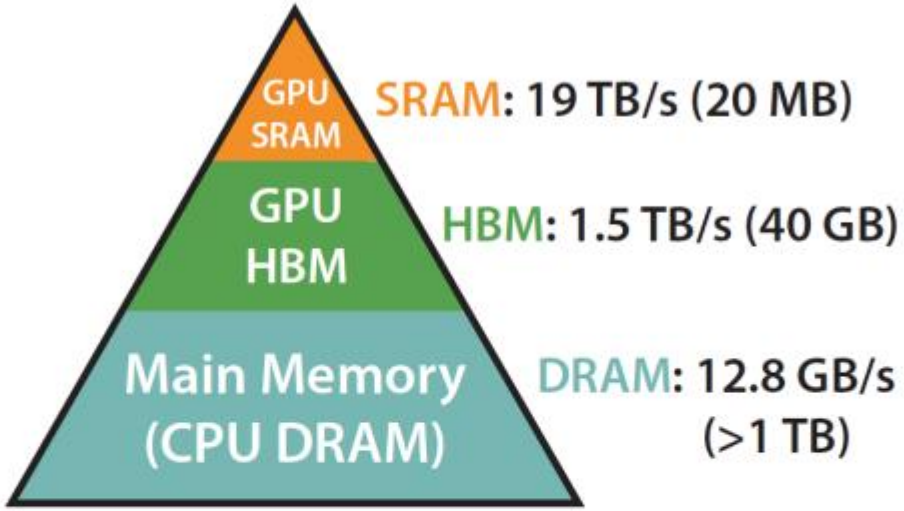
Inference is usually memory-bound

- Lots of time is wasted moving data around on the GPU instead of doing computation

Revisit: GPU Memory Hierarchy

Memory is arranged hierarchically

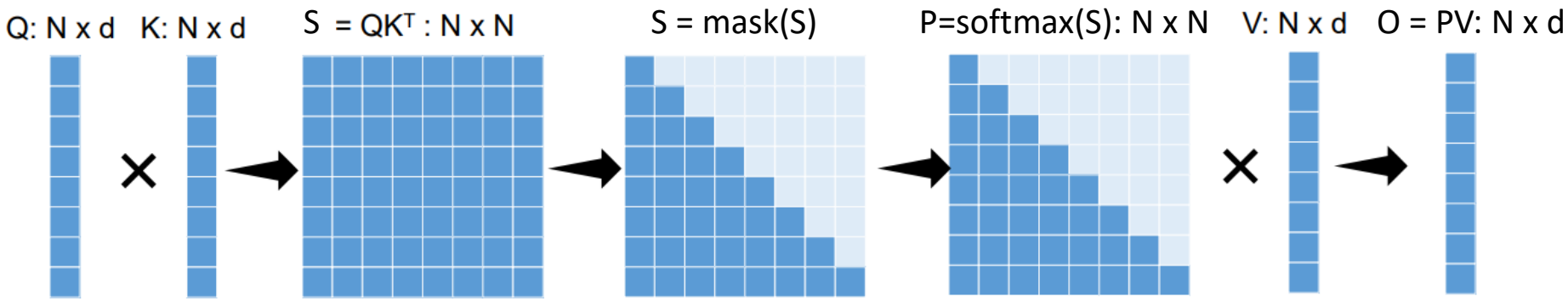
- GPU SRAM is small, and supports the fastest access
- GPU HBM is larger but with much slower access
- CPU DRAM is huge, but the slowest of all



Memory Hierarchy with Bandwidth & Memory Size

Standard Attention

Attention: $O = \text{Softmax}(QK^T)\,V$



$$\mathbf{S} = \mathbf{QK}^\top \in \mathbb{R}^{N\times N}, \quad \mathbf{P} = \text{softmax}(\mathbf{S}) \in \mathbb{R}^{N\times N}, \quad \mathbf{O} = \mathbf{PV} \in \mathbb{R}^{N\times d},$$

Algorithm 0 Standard Attention Implementation

Require: Matrices $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{N\times d}$ in HBM.

1: Load $\mathbf{Q}, \mathbf{K}$ by blocks from HBM, compute $\mathbf{S} = \mathbf{QK}^\top$, write $\mathbf{S}$ to HBM.
2: Read $\mathbf{S}$ from HBM, compute $\mathbf{P} = \text{softmax}(\mathbf{S})$, write $\mathbf{P}$ to HBM.
3: Load $\mathbf{P}$ and $\mathbf{V}$ by blocks from HBM, compute $\mathbf{O} = \mathbf{PV}$, write $\mathbf{O}$ to HBM.
4: Return $\mathbf{O}$.

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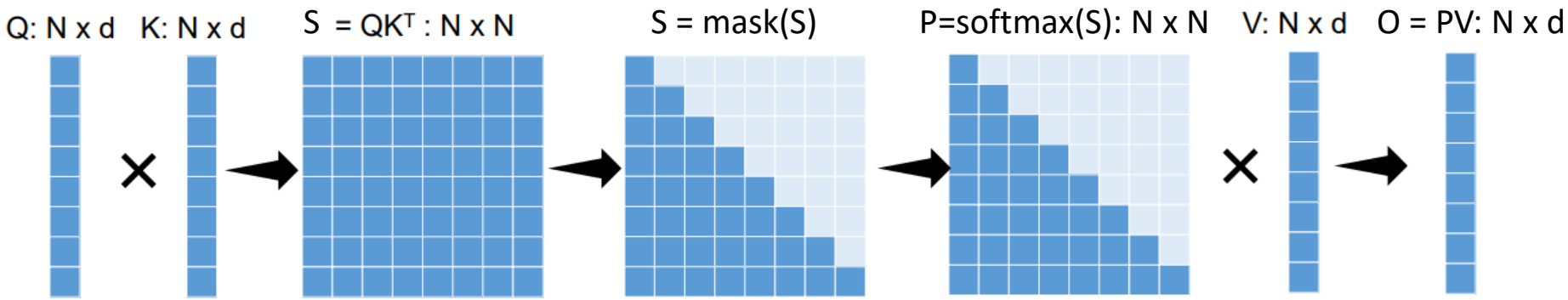

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Question: What are limitations of standard attention implementation?

Standard Attention

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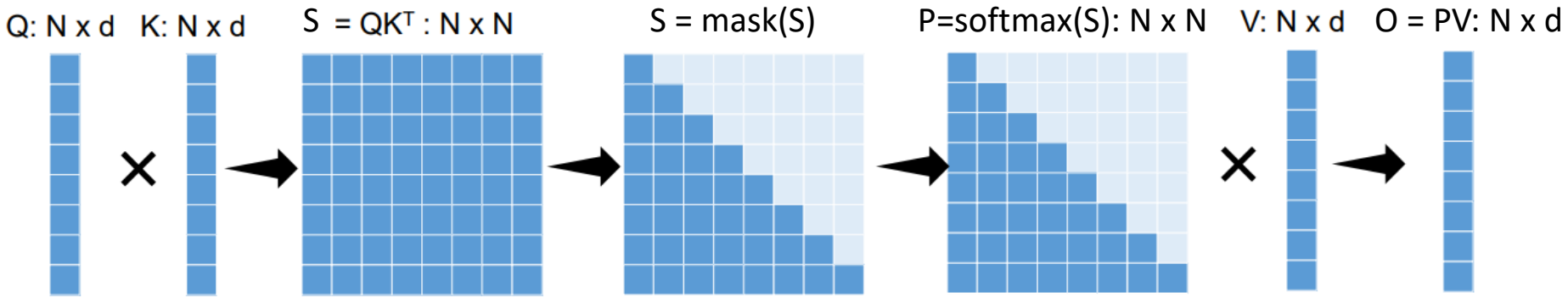

Challenges:

- Repeated reads/writes from GPU HBM
- Large intermediate results
- Cannot scale to long sequences due to O(N^2) intermediate results

FlashAttention

- Three key ideas are combined to obtain FlashAttention
  - Operator fusion: Use a single kernel that includes all operators during attention computation to avoid kernel launching overhead and intermediate data movement
  - Tiling: compute the attention weights block by block so that we don’t have to load everything into SRAM at once
  - Recomputation: don’t store the full attention matrix in forward, but just recompute during the backward pass

Operator Fusion

Version A: Usually, we compute a neural network one operator at a time by moving operation input to GPU SRAM (fast/small), doing some computation, then returning the output to GPU HBM (slow/large)

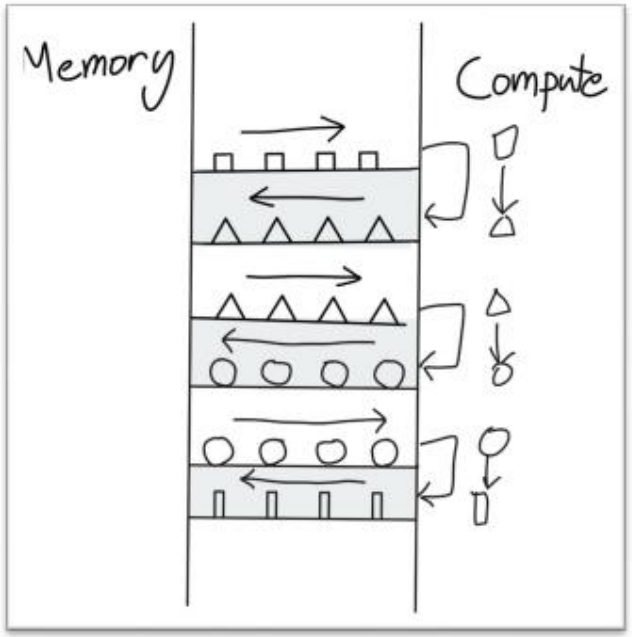

Figure from https://horace.io/brrr_intro.html

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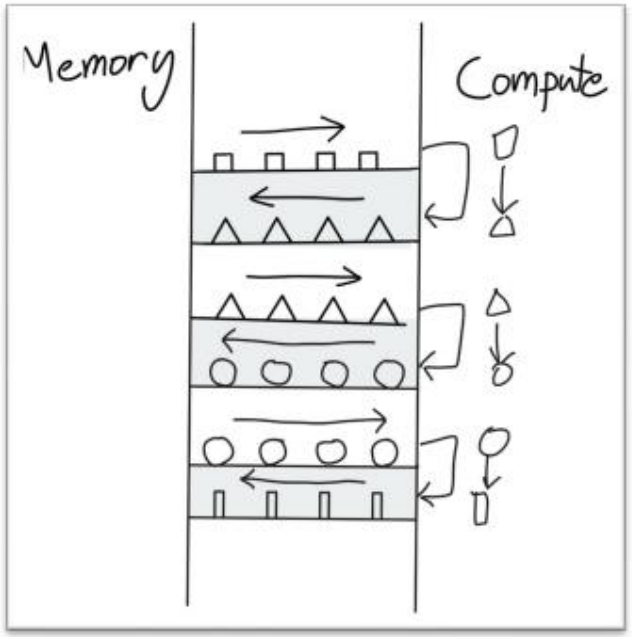

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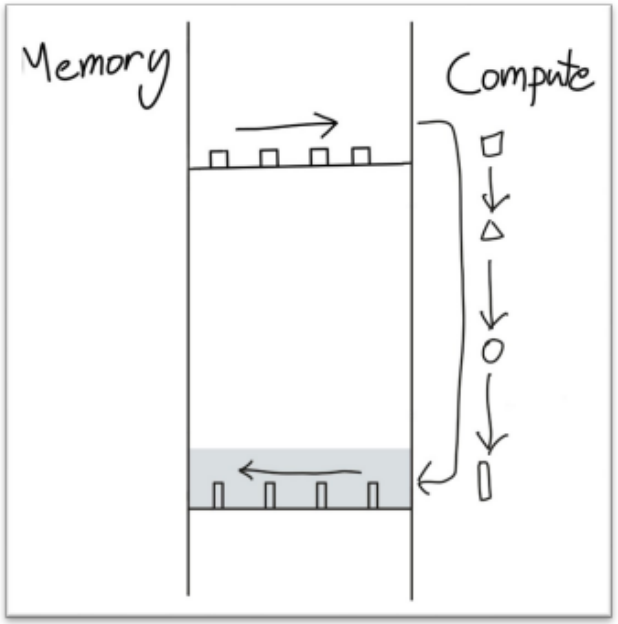
Version B: Operator fusion instead moves the original input to GPU SRAM (fast/small), does a whole sequence of layer computations without ever touching HBM, and then returns the final layer output to GPU HBM (slow/large)



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Operator Fusion

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Version B: Operator fusion instead moves the original input to GPU SRAM (fast/small), does a whole sequence of layer computations without ever touching HBM, and then returns the final layer output to GPU HBM (slow/large)

Version A is how standard attention is implemented

$$\mathbf{S} = \mathbf{Q}\mathbf{K}^\top \in \mathbb{R}^{N \times N}, \quad \mathbf{P} = \mathrm{softmax}(\mathbf{S}) \in \mathbb{R}^{N \times N}, \quad \mathbf{O} = \mathbf{P}\mathbf{V} \in \mathbb{R}^{N \times d},$$

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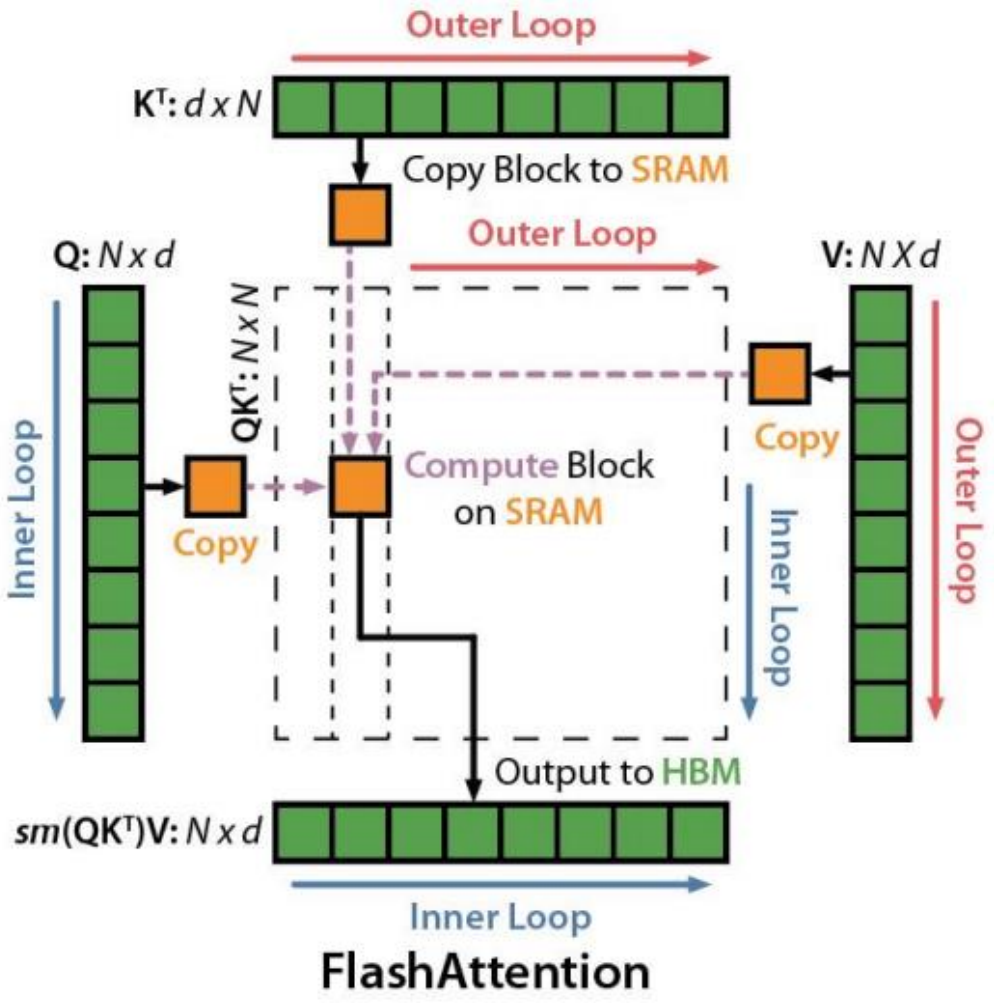
Tiling: Decompose Large Attention Calculation into Smaller Blocks

1. Load inputs by blocks from global to shared memory
2. On chip, compute attention output wrt the block
3. Update output in device memory by scaling



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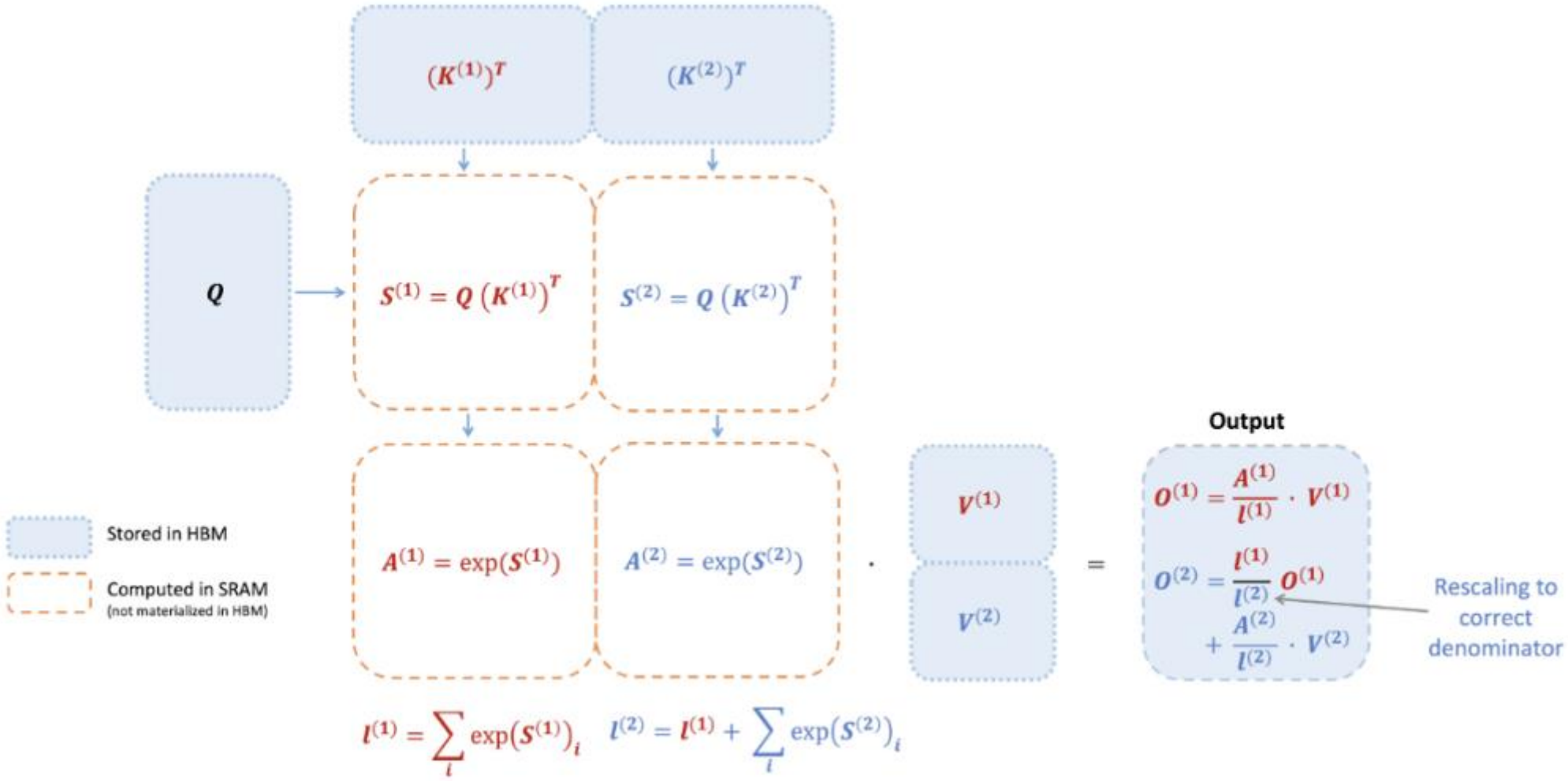

Figure from http://arxiv.org/abs/2307.08691

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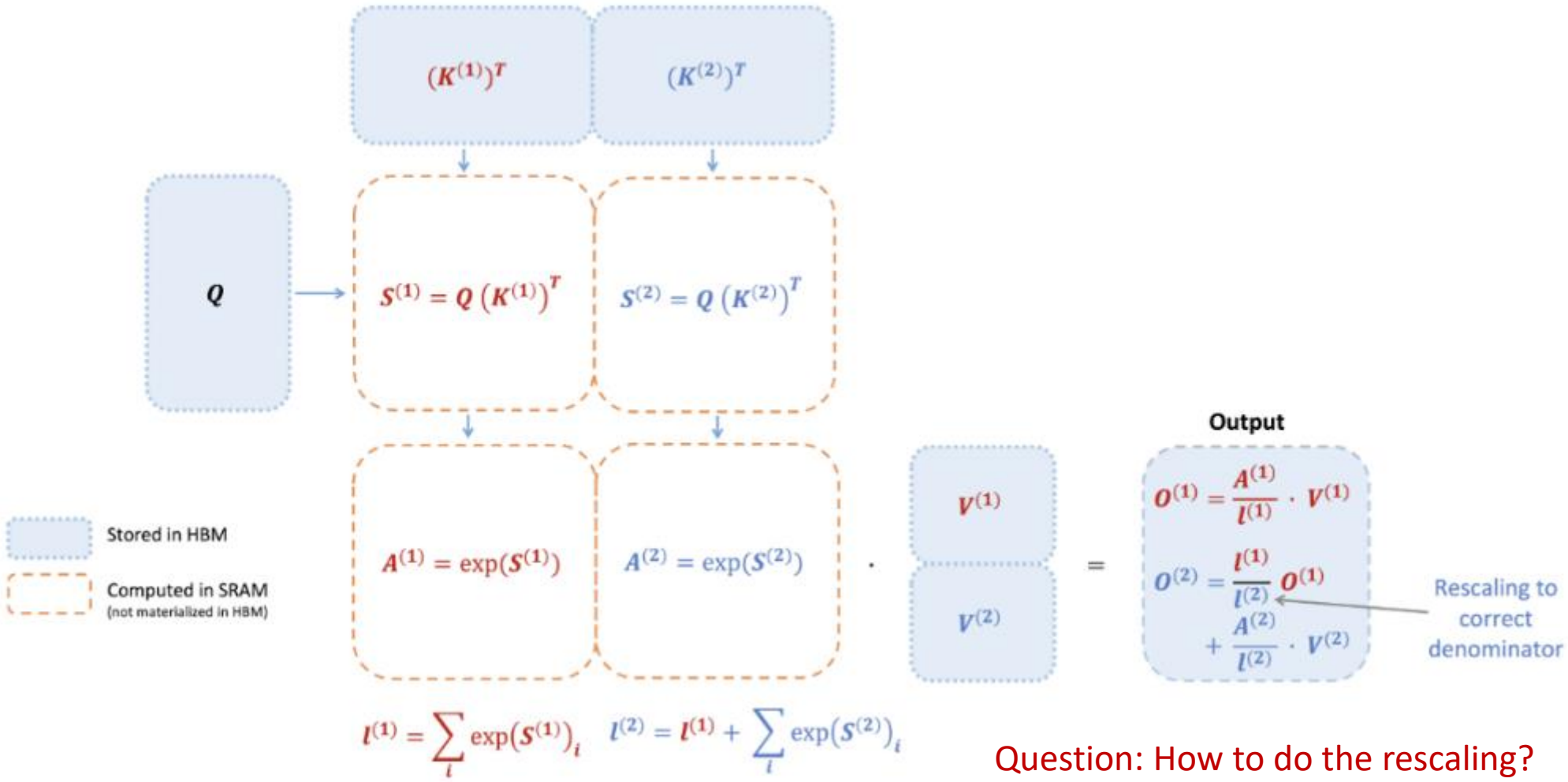

Question: How to do the rescaling?

Figure from http://arxiv.org/abs/2307.08691

How to Implement Softmax? Standard Softmax

For a vector $x \in \mathbb{R}^B$, softmax is computed as:

$$\text{softmax}(x) = \frac{e^{x_i}}{\sum_j e^{x_j}}$$

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Issue: Easily lead to numerical instability, e.g., overflow because of $\sum_j e^{x_j}$

Stable Softmax

Subtracting the max value from the input vector before applying the exp function, which helps prevent overflow issue

1. Compute the maximum value in x:

$$m(x) := \max_i x_i$$

2. Compute the adjusted exponentials:

$$f(x) := \left[e^{x_1 - m(x)}, e^{x_2 - m(x)}, \ldots, e^{x_B - m(x)}\right]$$

3. Compute the normalization denominator:

$$\ell(x) := \sum_i f(x)_i$$

4. Compute the final softmax:

$$\text{softmax}(x) = \frac{f(x)}{\ell(x)}$$

FlashAttention Tiled Softmax: Tiling + Online Softmax

Let's say we have two blocks $x^{(1)}$ and $x^{(2)}$ each of size B. The concatenated vector is:

$$x = [x^{(1)}, x^{(2)}] \in \mathbb{R}^{2B}$$

1. Track the max value across blocks

$$m(x) = \max(m(x^{(1)}), m(x^{(2)}))$$

2. Compute adjusted exponentials:

$$f(x) = \left[e^{m(x^{(1)})-m(x)} f(x^{(1)}), e^{m(x^{(2)})-m(x)} f(x^{(2)})\right]$$

3. Compute the normalization denominator (incrementally):

$$\ell(x) = e^{m(x^{(1)})-m(x)} \ell(x^{(1)}) + e^{m(x^{(2)})-m(x)} \ell(x^{(2)})$$

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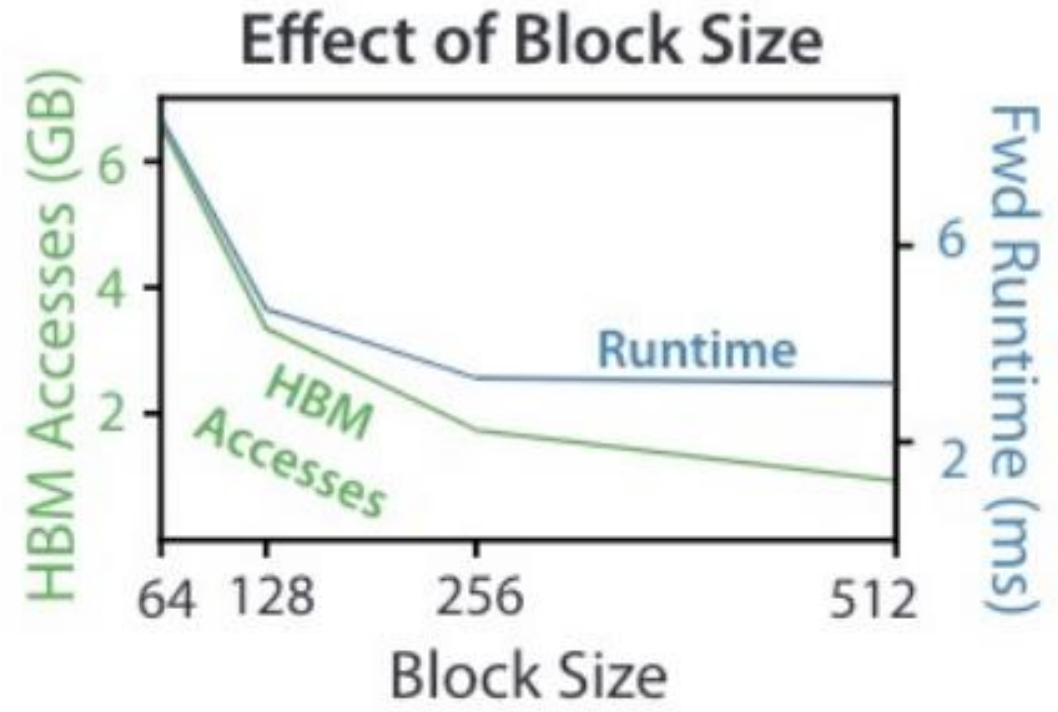
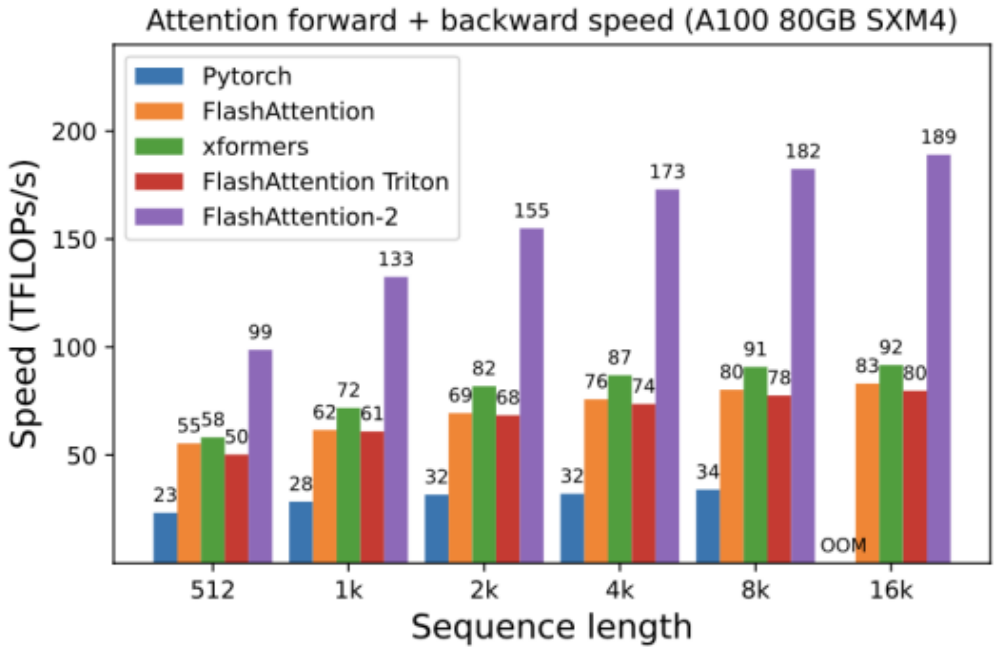
Only need to track intermediate statistics (m(x_i), l(x_i)) to compute softmax one block at a time

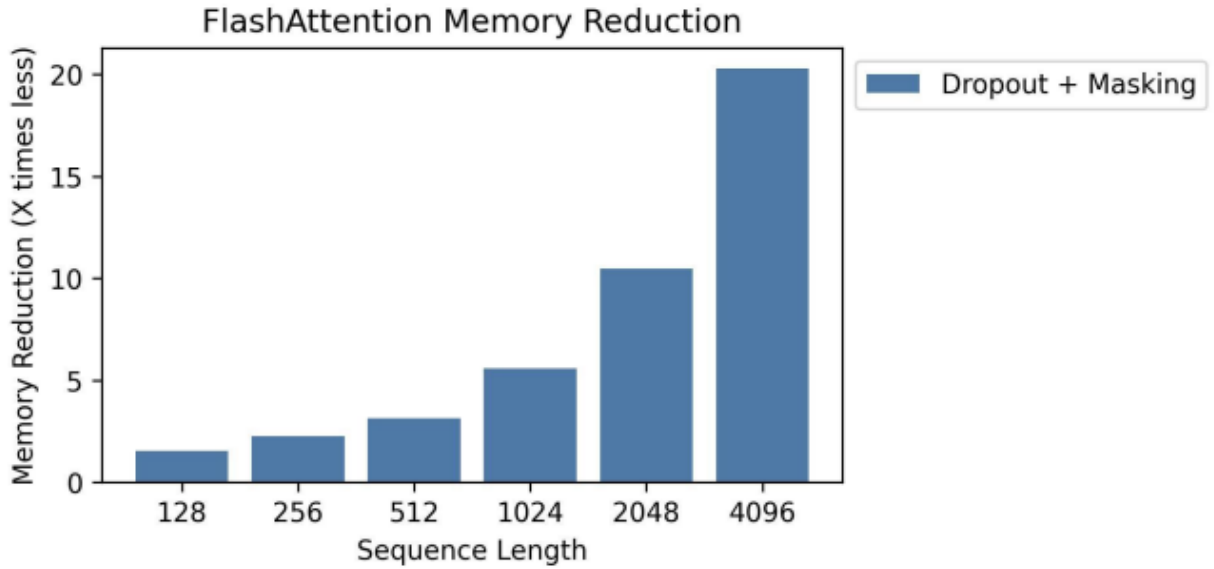
FlashAttention Results

The algorithm is performing exact attention, no reduction in perplexity or quality of the model



FlashAttention: 2-4x speedup, 10-20x memory reduction





Memory linear in sequence length

Is FlashAttention Stable?

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Alicia Golden[1,2] Samuel Hsia[1,2] Fei Sun[3] Bilge Acun[1] Basil Hosmer[1] Yejin Lee[1]
Zachary DeVito[1] Jeff Johnson[1] Gu-Yeon Wei[2] David Brooks[2] Carole-Jean Wu[1]

[1]FAIR at Meta [2]Harvard University [3]Meta

***Abstract*—Training large-scale machine learning models poses distinct system challenges, given both the size and complexity of today's workloads. Recently, many organizations training state-of-the-art Generative AI models have reported cases of instability during training, often taking the form of loss spikes. Numeric deviation has emerged as a potential cause of this training instability, although quantifying this is especially challenging given the costly nature of training runs. In this work, we develop a principled approach to understanding the effects of numeric deviation, and construct proxies to put observations into context when downstream effects are difficult to quantify. As a case study, we apply this framework to analyze the widely-adopted Flash Attention optimization. We find that Flash Attention sees roughly an order of magnitude more numeric deviation as compared to Baseline Attention at BF16 when measured during an isolated forward pass. We then use a data-driven analysis based on the Wasserstein Distance to provide upper bounds on how this numeric deviation impacts model weights during training, finding that the numerical deviation present in Flash Attention is 2-5 times less significant than low-precision training.**

***Index Terms*—Generative AI, Numeric Deviation, Training Instability, Attention, Transformers**

One under-explored potential cause of training instability is *numeric deviation*. Numeric deviation between an optimization and its corresponding baseline can lead to the gradual accumulation of errors, which over the course of training have the potential to culminate in loss spikes that require a resetting of the model state [1]. This is challenging to quantify, as training's stochastic nature suggests some level of numeric deviation might be acceptable, yet determining the threshold for when training becomes unstable proves difficult.

In this work, we develop a principled quantitative approach to understanding numeric deviation in training optimizations. Our approach consists of two phases, including (i) developing a microbenchmark to perturb numeric precision in the given optimization, and (ii) evaluating how numeric deviation translates to changes in model weights through a data-driven analysis based on Wasserstein distance. This ultimately allows us to provide an upper bound on the amount of numeric deviation for a given optimization, and helps to contextualize the improvement within known techniques. We aim to use

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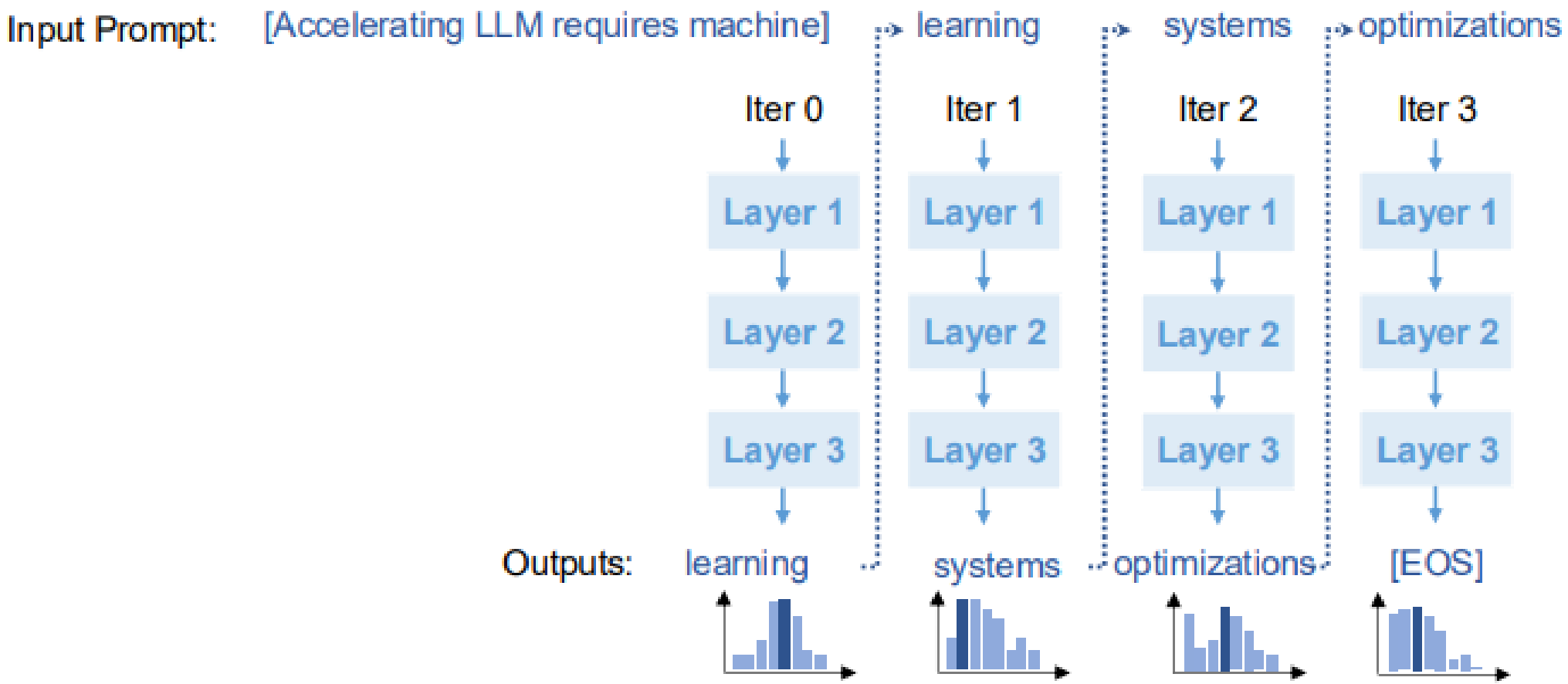
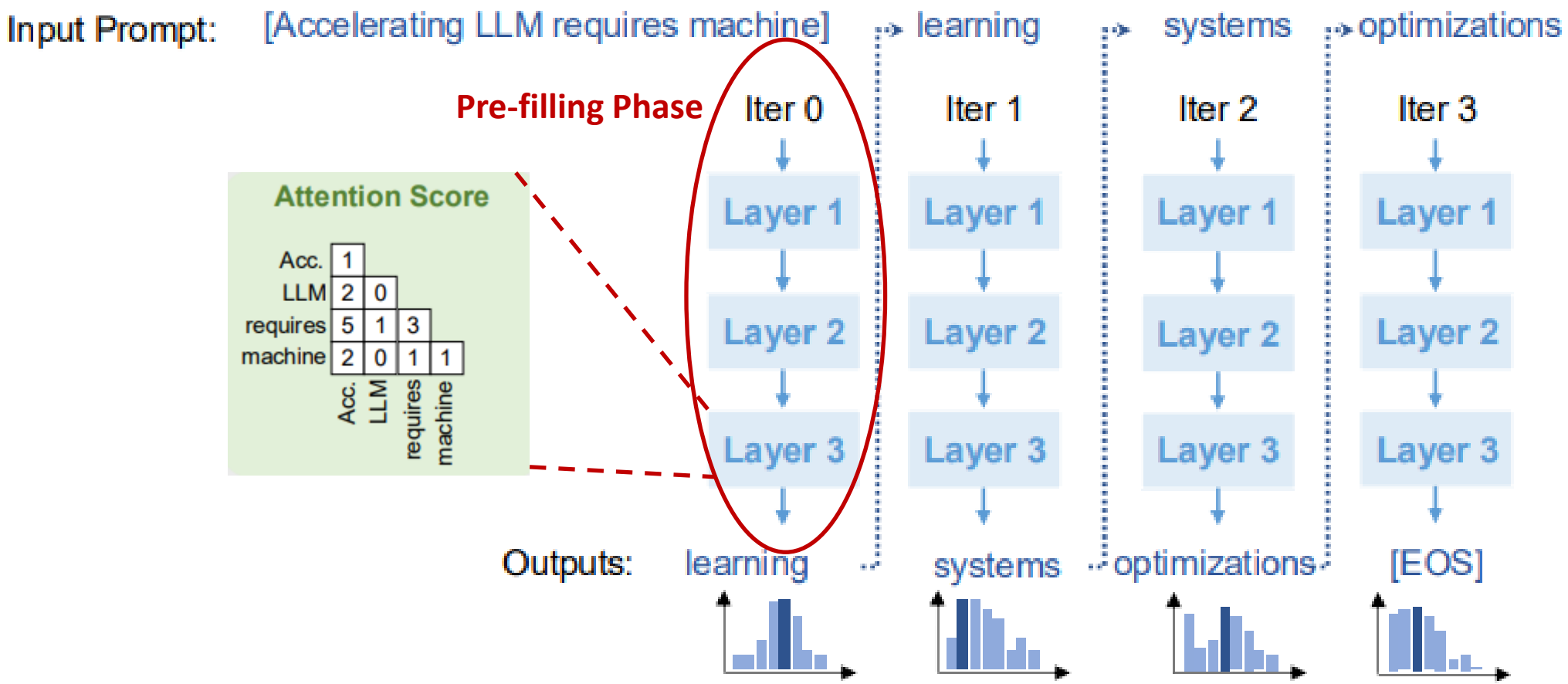
Generative LLM Inference: Autoregressive Decoding



Repeat until the sequence

- Reaches its pre-defined maximum length (e.g., 2048 tokens)
- Generates certain tokens (e.g., "<|*end of sequence*|>")

Generative LLM Inference: Autoregressive Decoding



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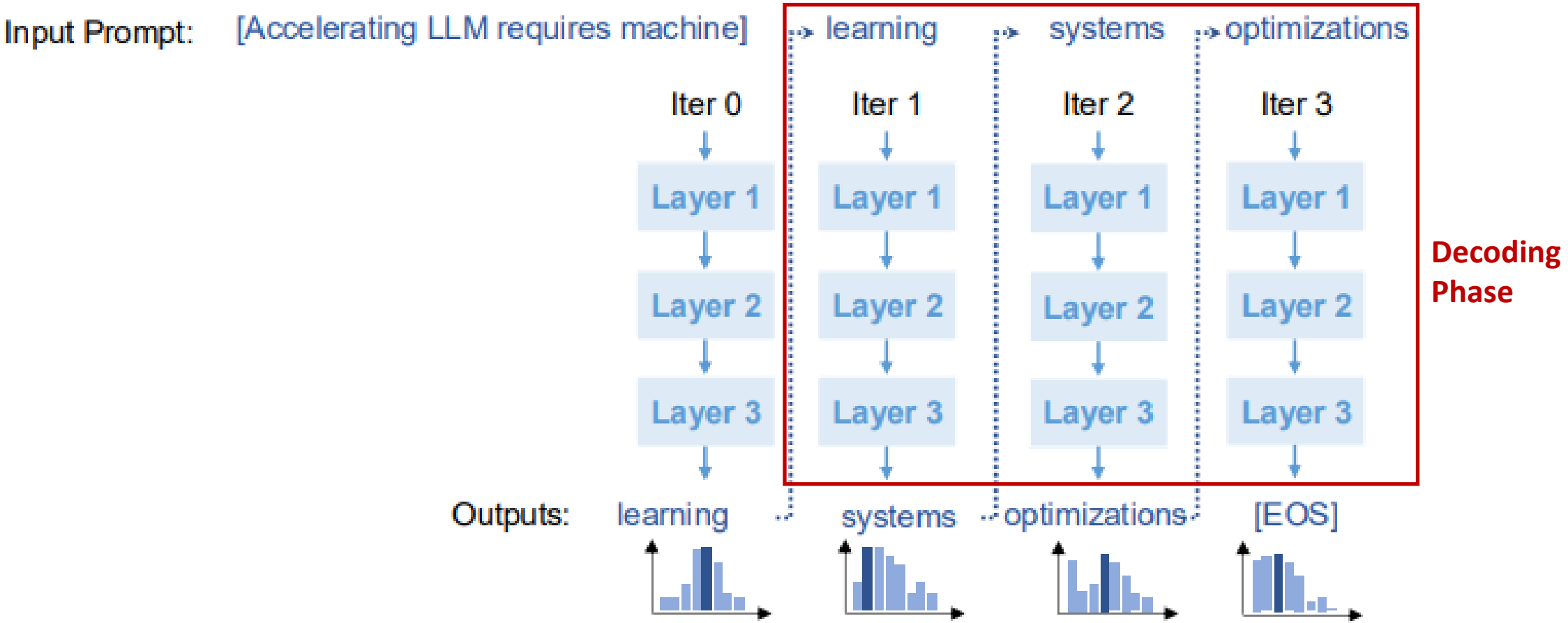
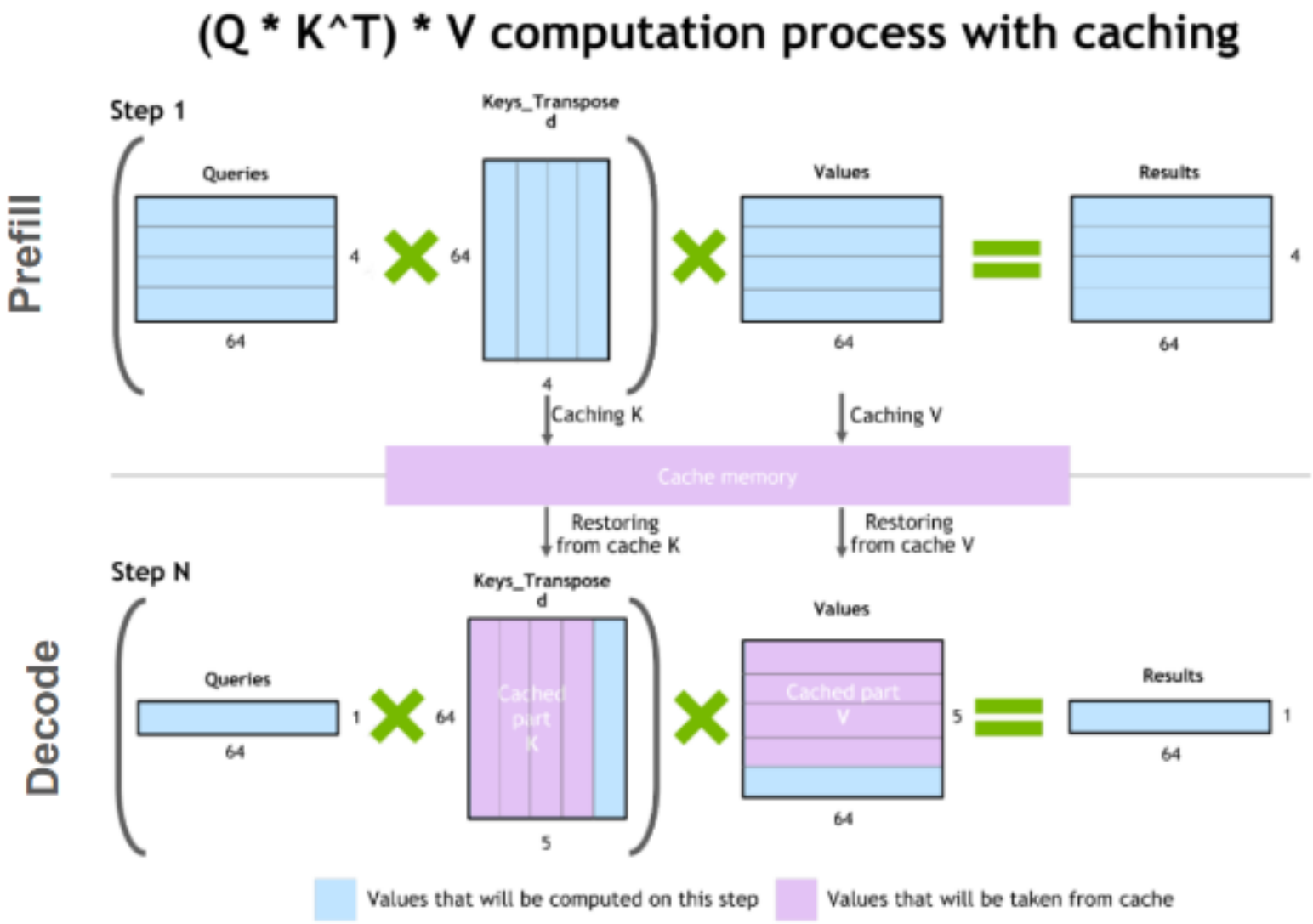


Generative LLM Inference: Autoregressive Decoding

- **Pre-filling phase** (0-th iteration):
  - Process ***all*** input tokens at once
- **Decoding phase** (all other iterations):
  - Process a ***single*** token generated from previous iteration
  - Use attention keys & values of all previous tokens

- Key-value cache:
  - Save attention keys and values for the following iterations to avoid recomputation

Generative LLM Inference: KV Cache



Generative LLM Inference: Important Metrics

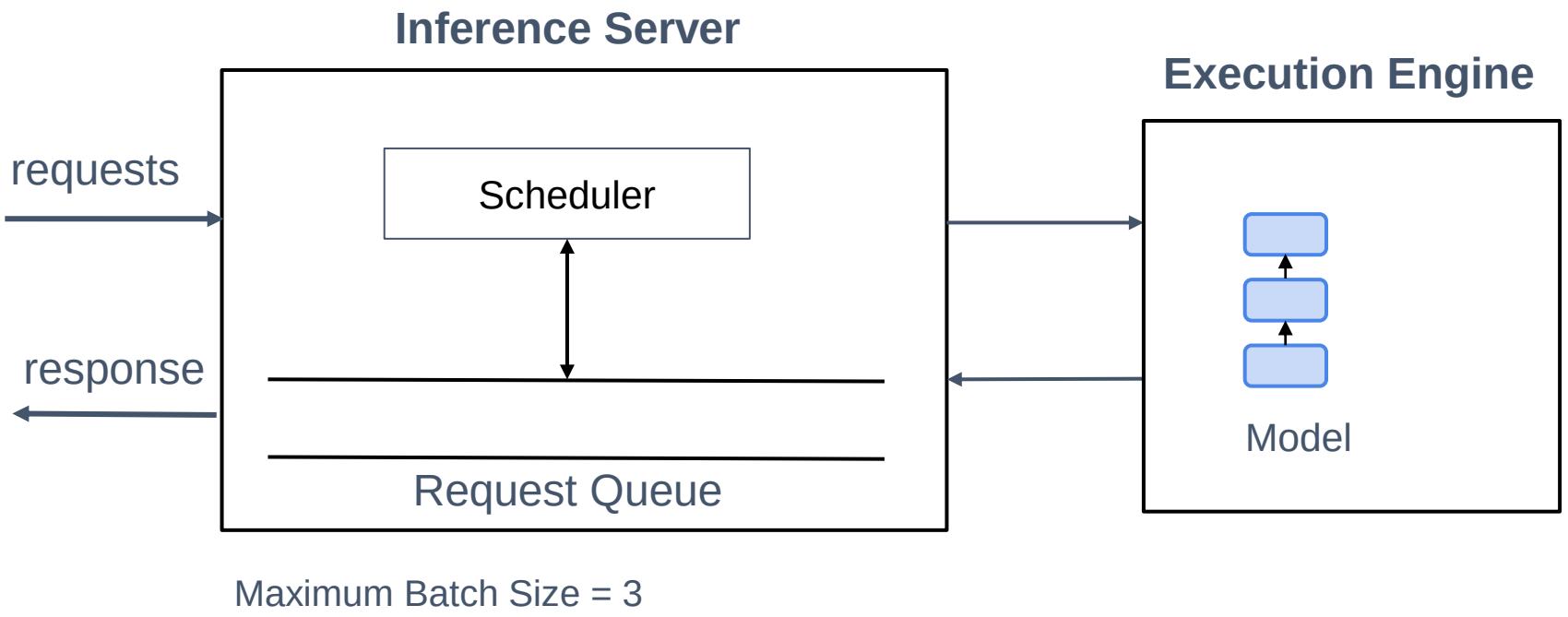
- **Time To First Token (TTFT):** Measures how quickly users begin to see model output after submitting a query.
  - Crucial for real-time interactions.
  - Driven by prompt processing time and the generation of the first token.
- **Time Per Output Token (TPOT):** Time taken to generate each output token.
  - Impacts user perception of speed *(e.g., 100 ms/token = 10 tokens/second, ~450 words/minute).*
- **Latency = (TTFT) + (TPOT * the number of tokens to be generated).**
  - Total time to generate the complete response.
- **Throughput:** Number of tokens generated per second across all requests by the inference server.

LLM Inference Performance Engineering: Best Practices | Databricks Blog

Optimizing LLM Inference: Goals and Tradeoffs

- **Optimization Goals & Tradeoffs:**
  - **Goal:** *Minimize* TTFT, *maximize* throughput, and *reduce* TPOT.
  - **Throughput vs. TPOT Tradeoff:** *Processing multiple queries concurrently increases throughput but extends TPOT for each user.*
- **Key Heuristics for Model Evaluation:**
  - **Output Length:** Dominates latency.
  - **Input Length:** Minimal impact on performance but significant for hardware.
  - **Model Size:** Larger models have higher latency, but not proportional to their size.
    - Example: Llama-70B is 2x Llama-13B.

LLM Serving System



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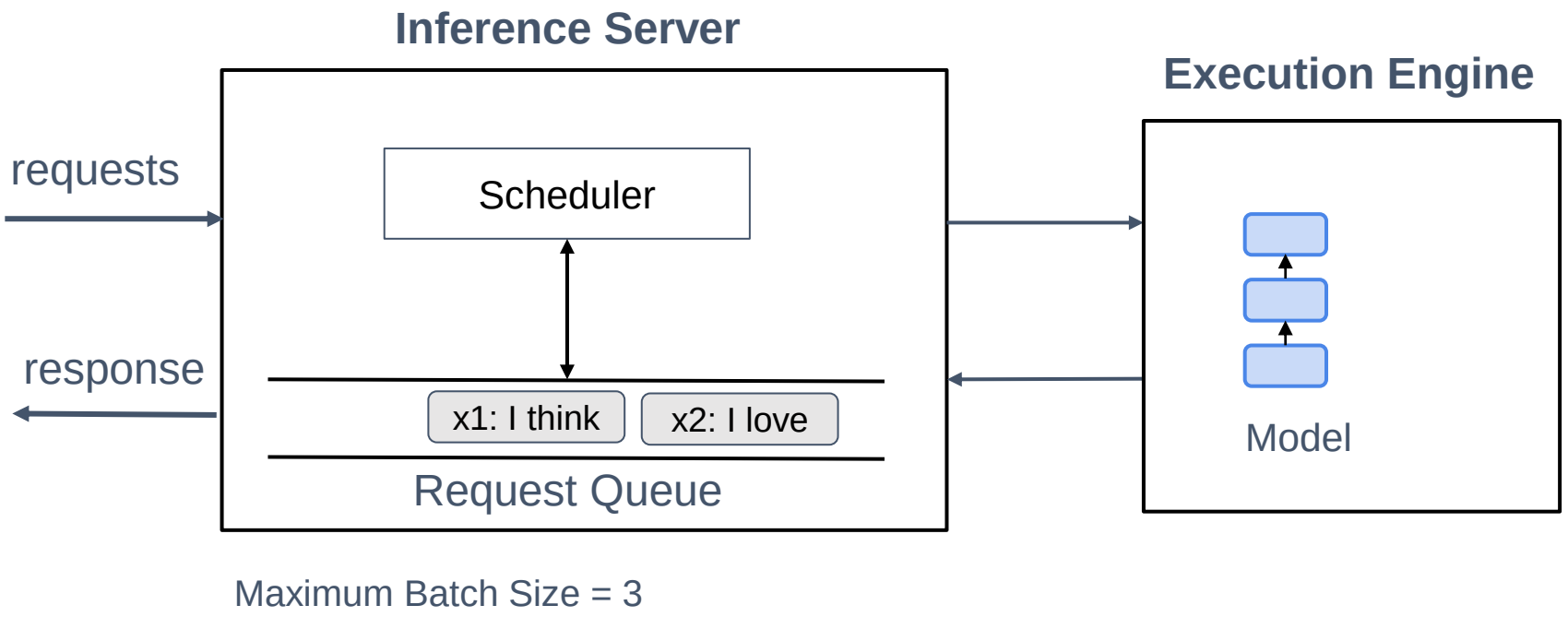

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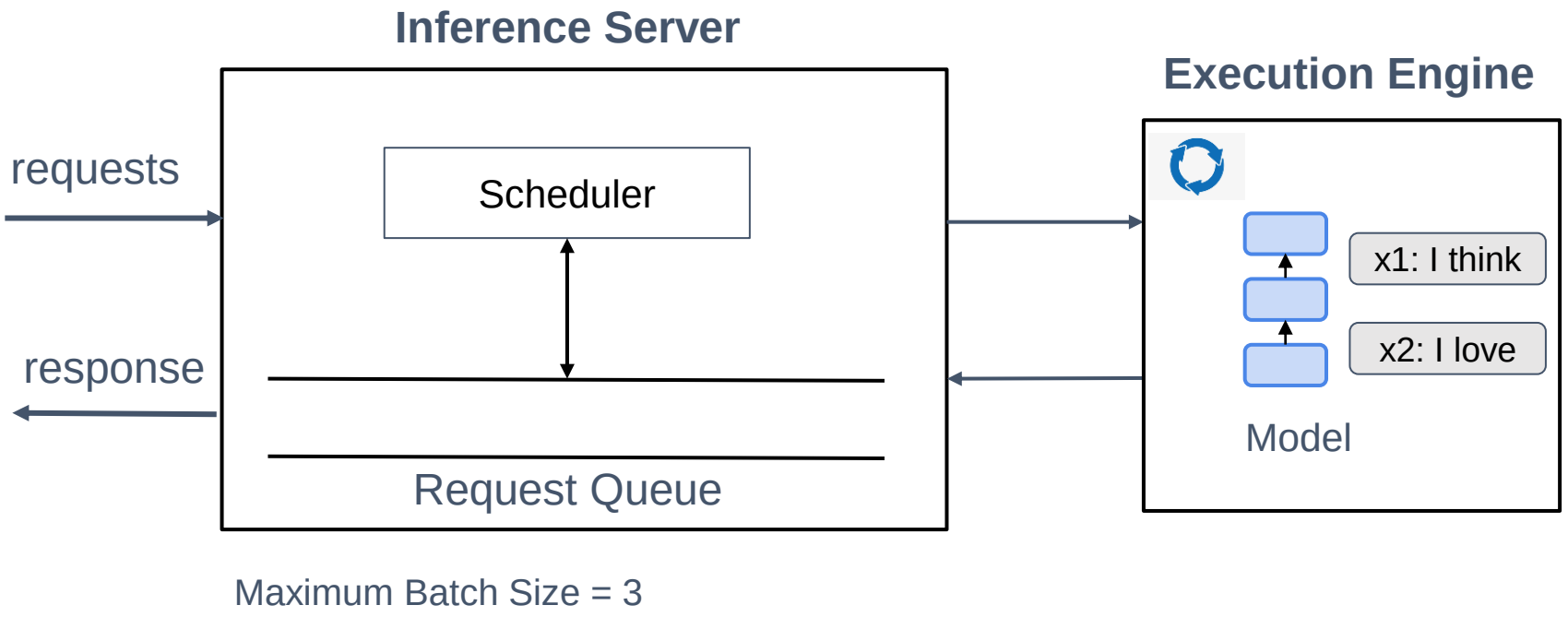

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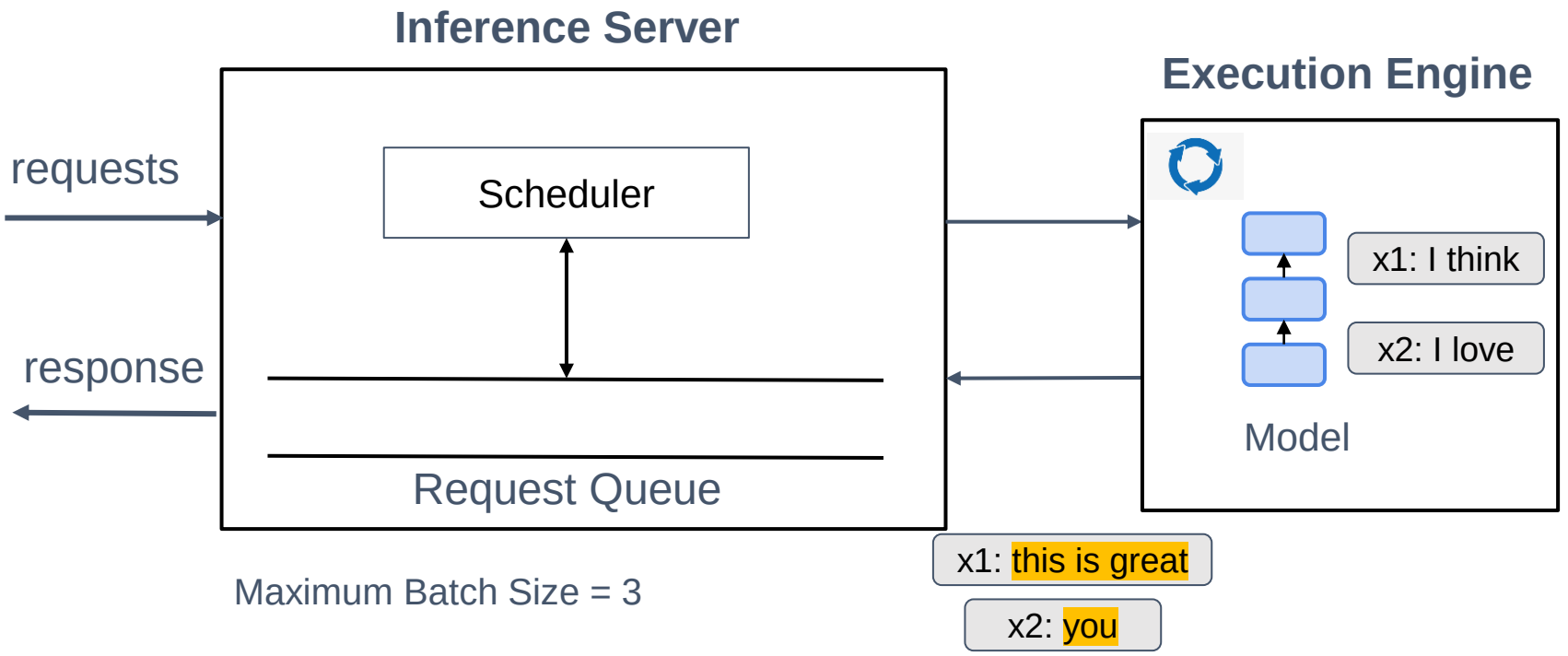

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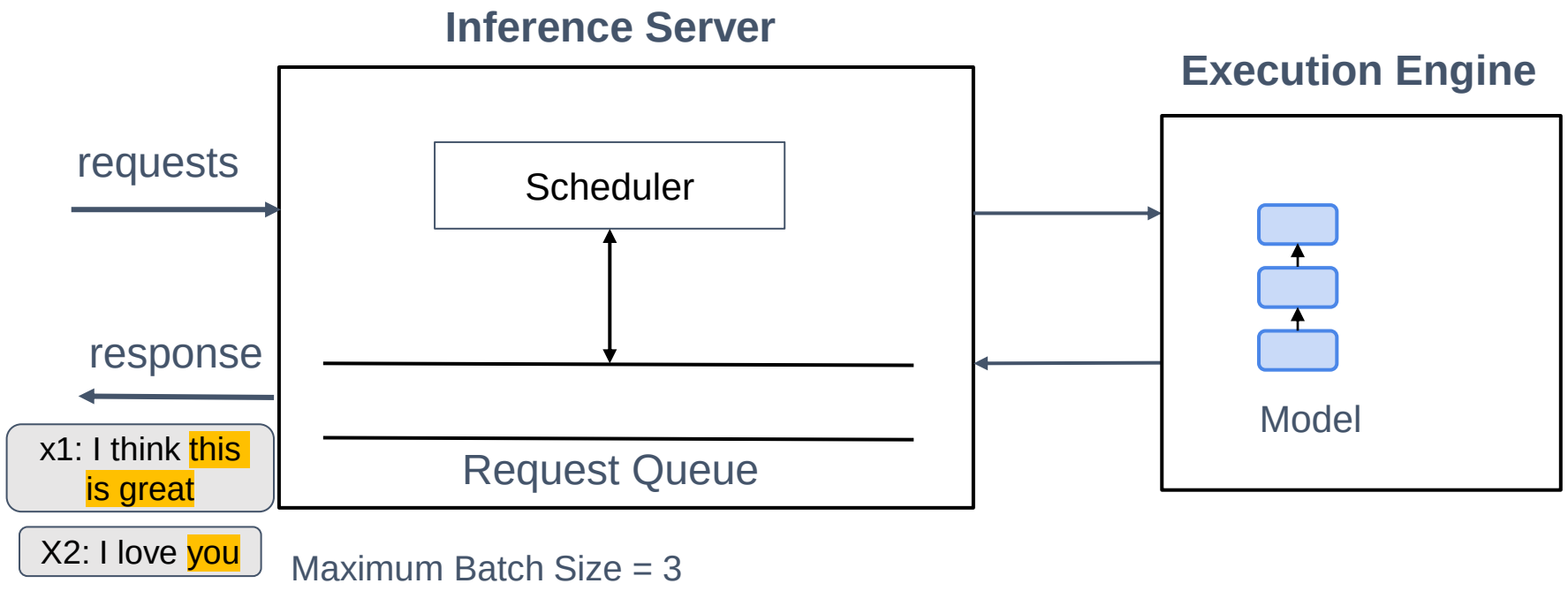
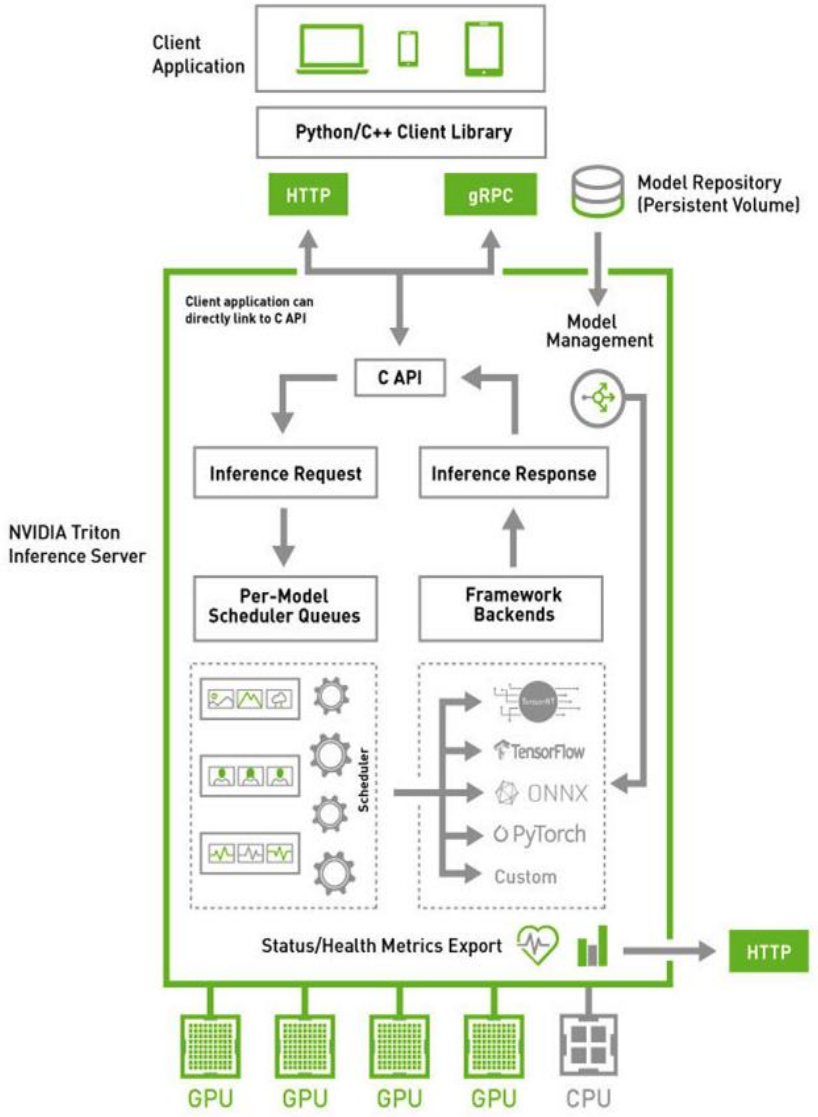
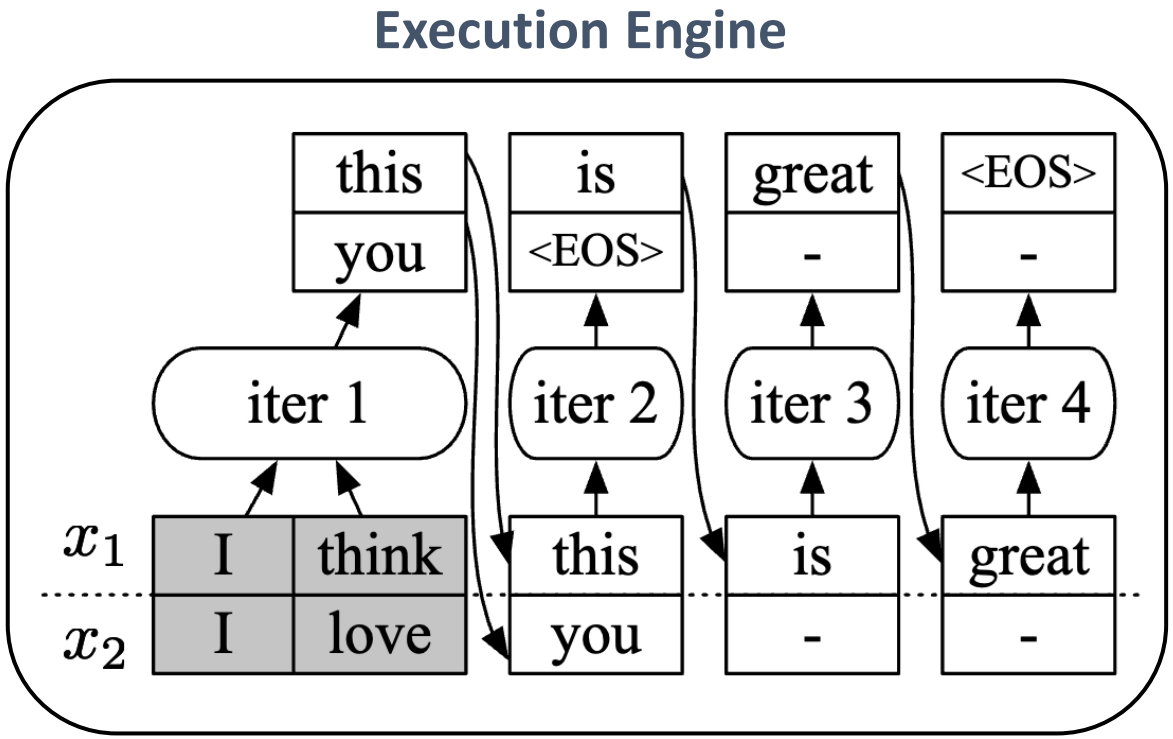


Triton Inference Server / TensorFlow Serving

- Separates implementation of serving layer and execution layer
- Implements scheduling and batching algorithms
  - Dynamic Batching
  - Sequence Batching
- Allows multiple models to concurrently execute
- Supports multiple frameworks
  - TensorRT
  - TensorFlow
  - PyTorch
  - ONNX



Problem 1: Request Level Scheduling



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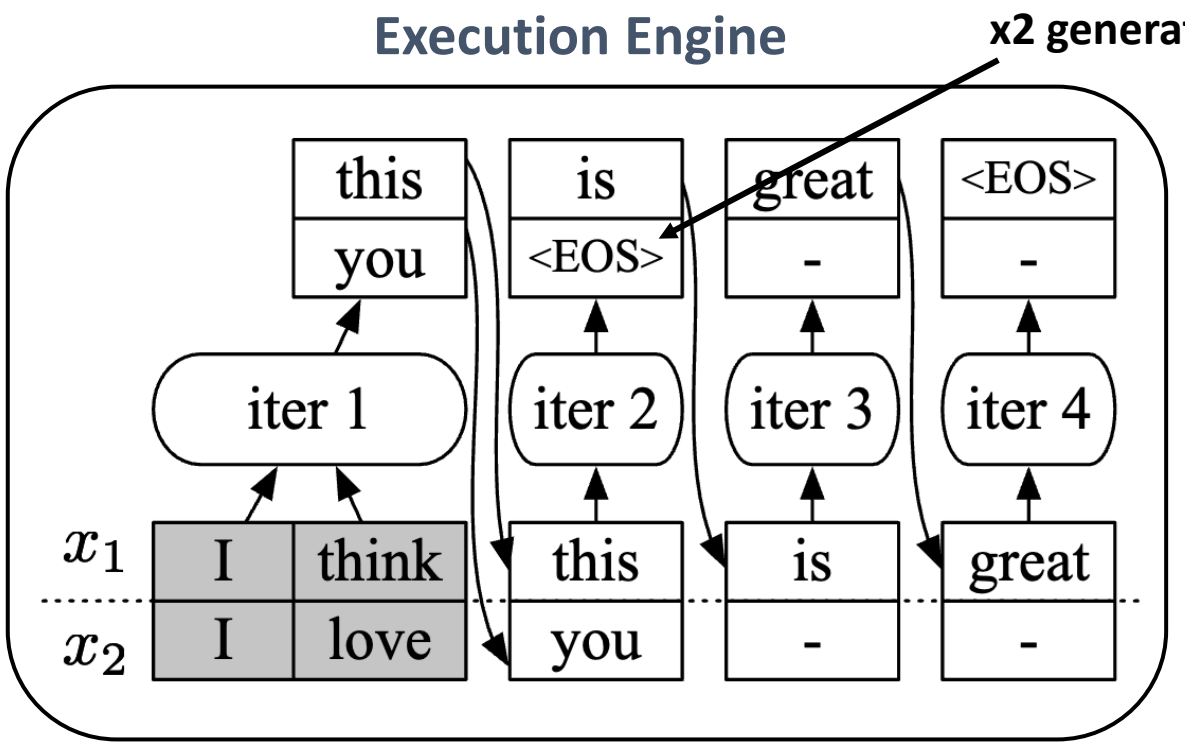
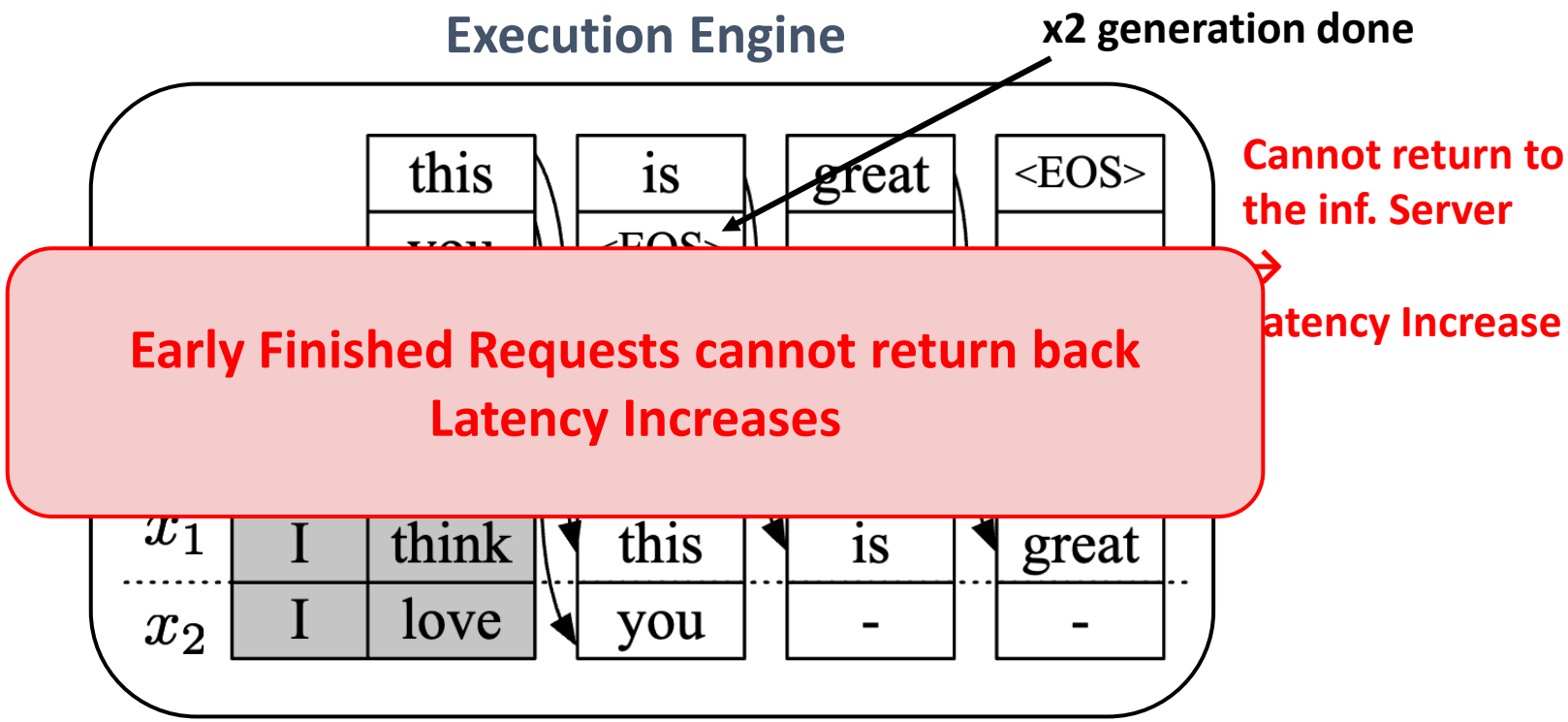


Cannot return to the inf. Server
→
Latency Increase

Problem 1: Request Level Scheduling



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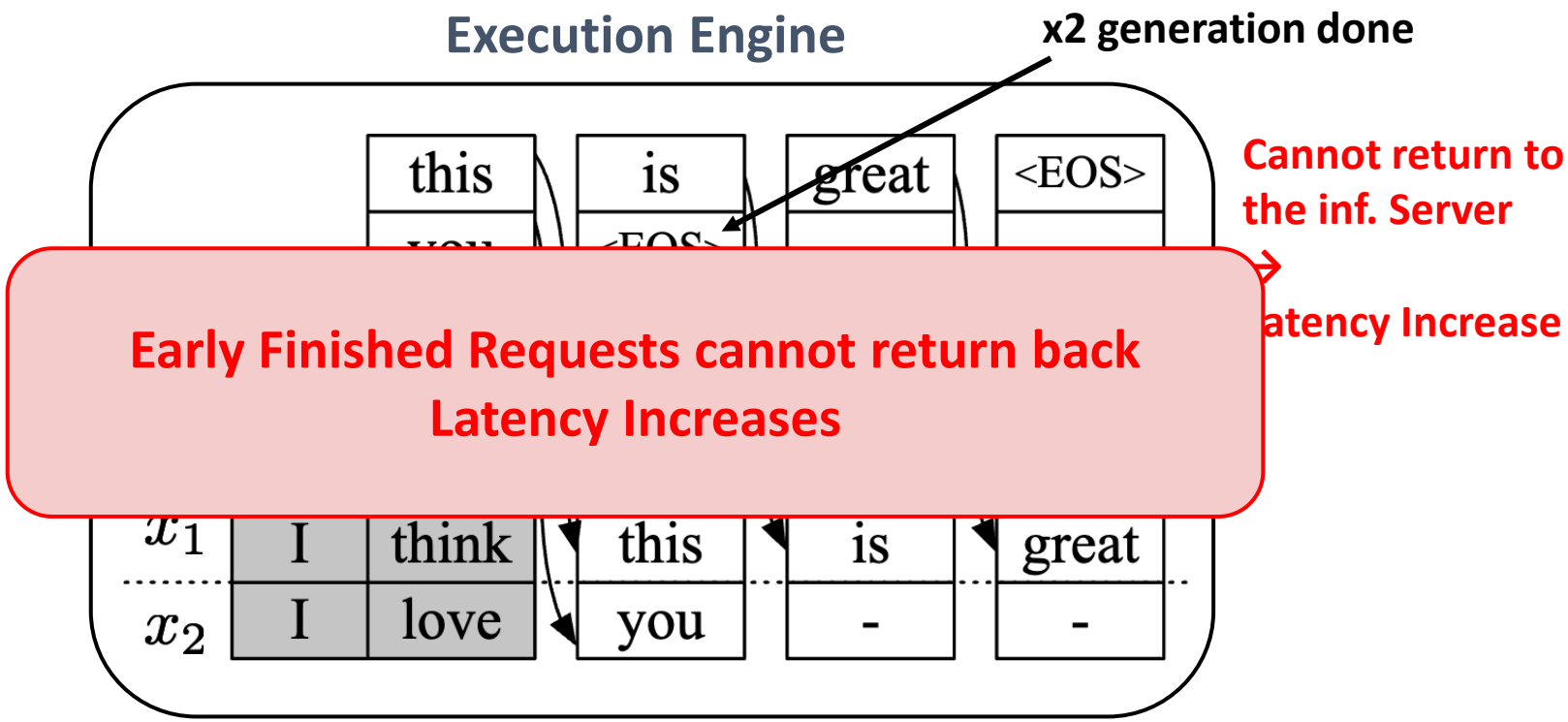

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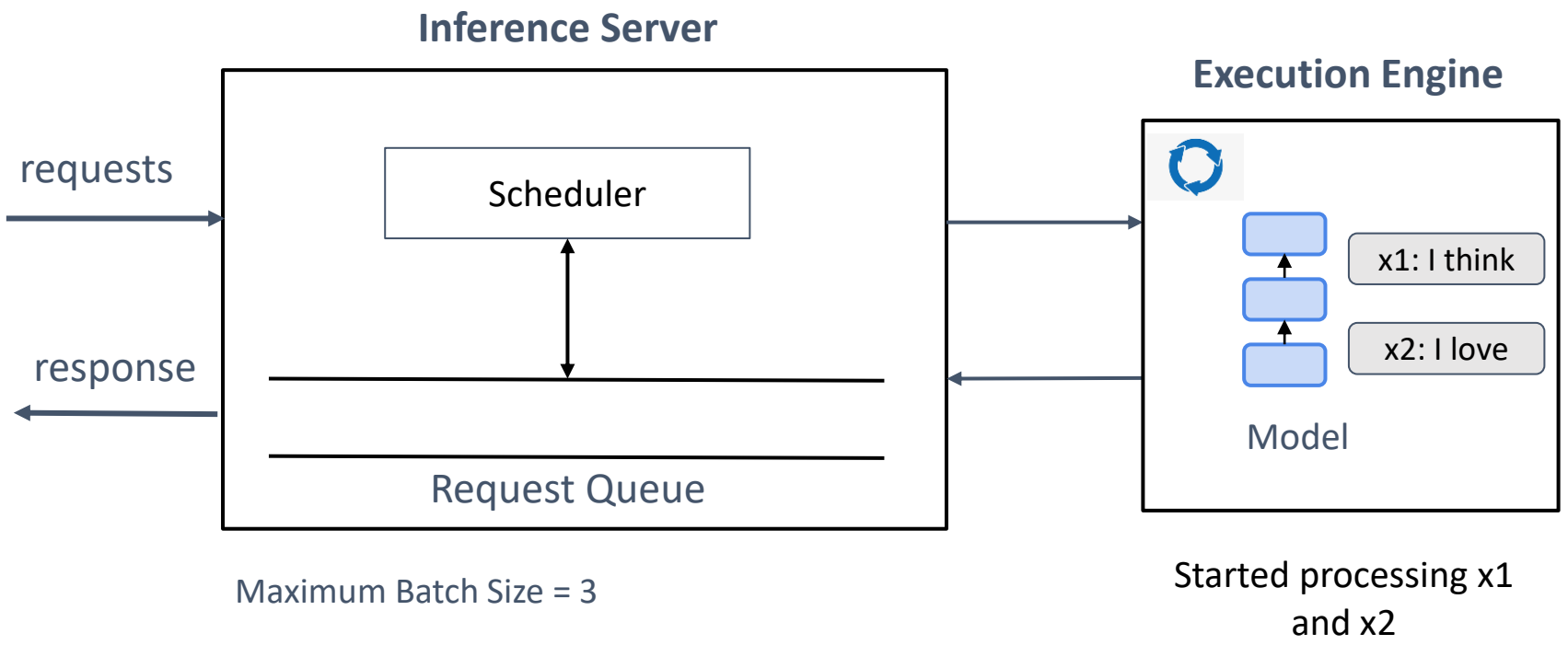

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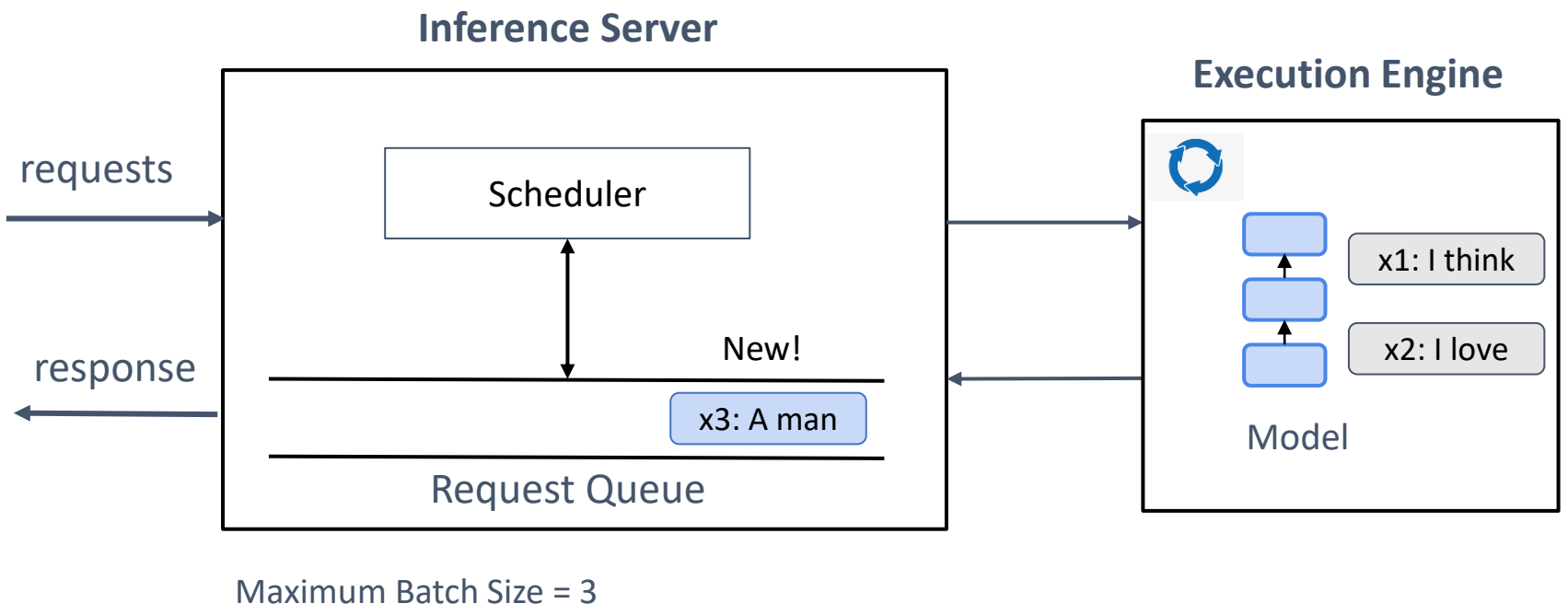

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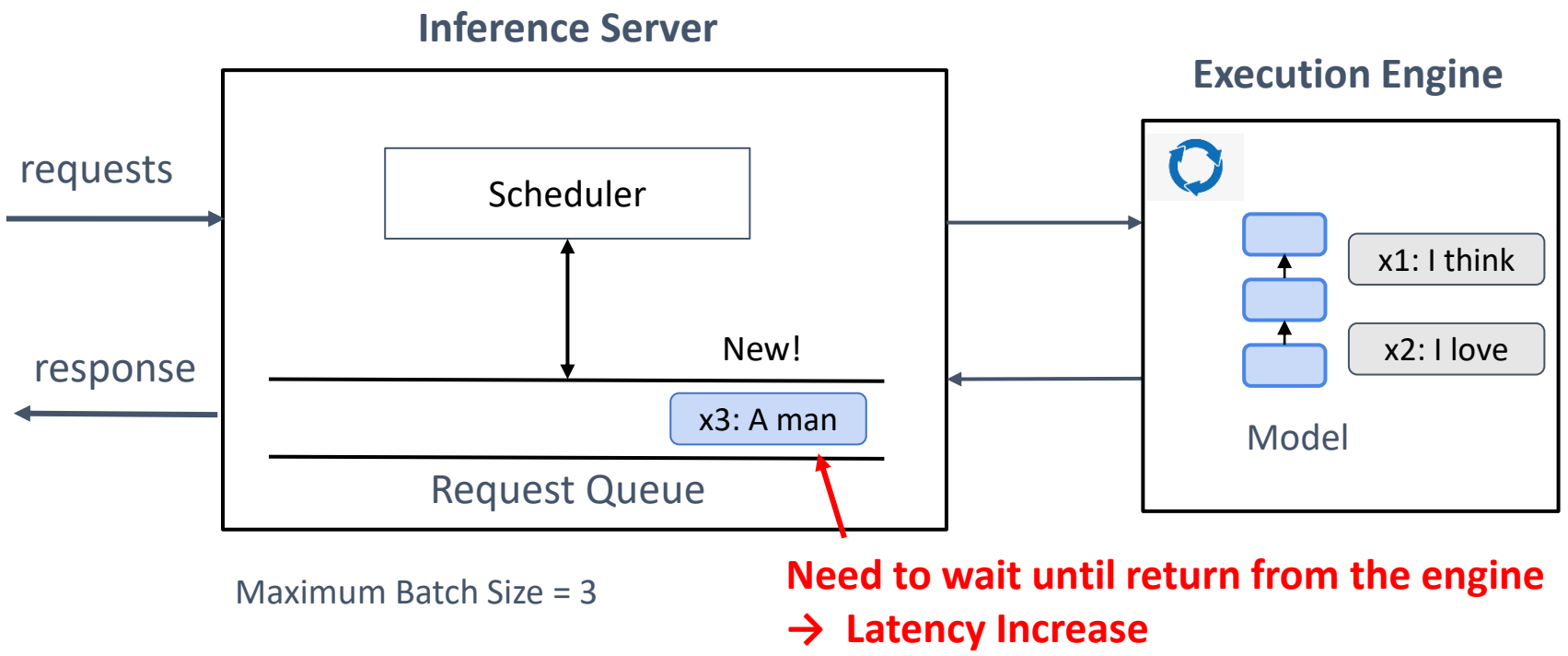

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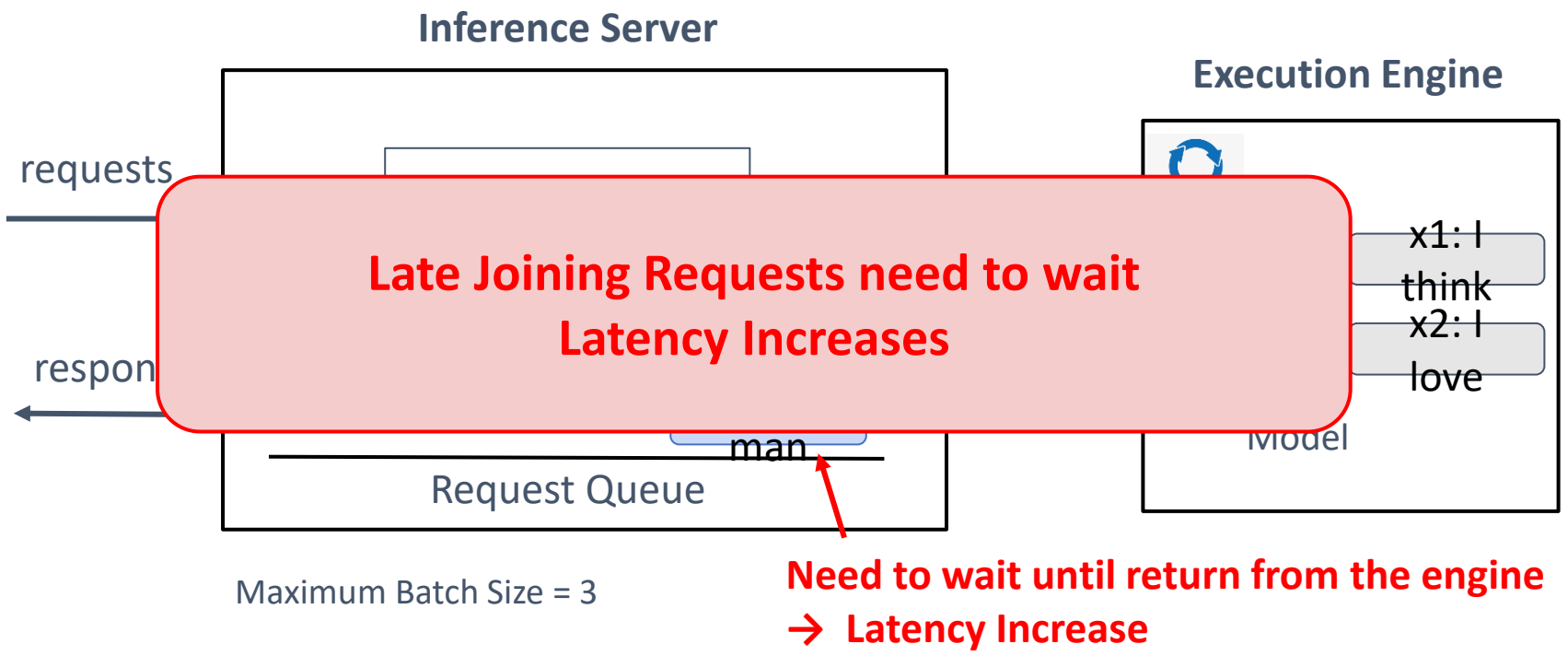
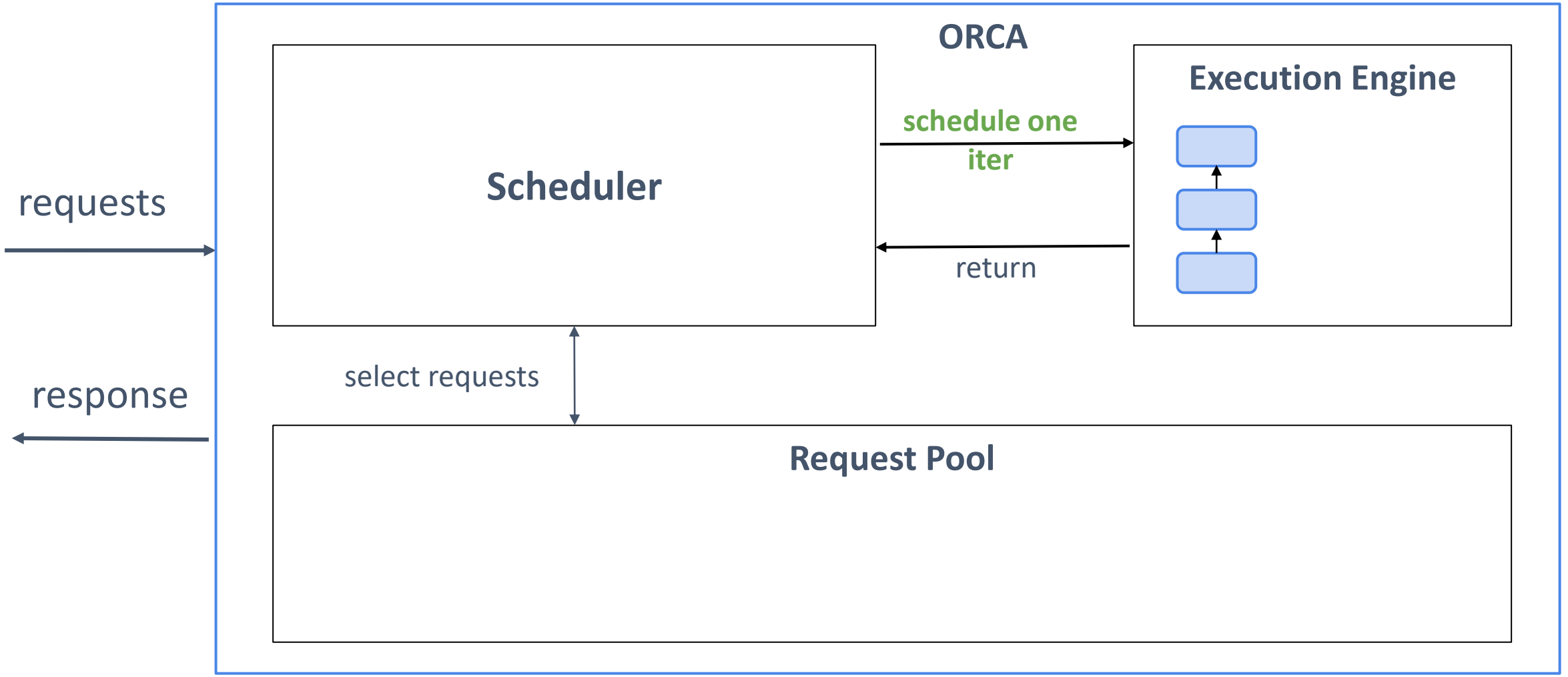


Solution 1: Iteration Level Scheduling



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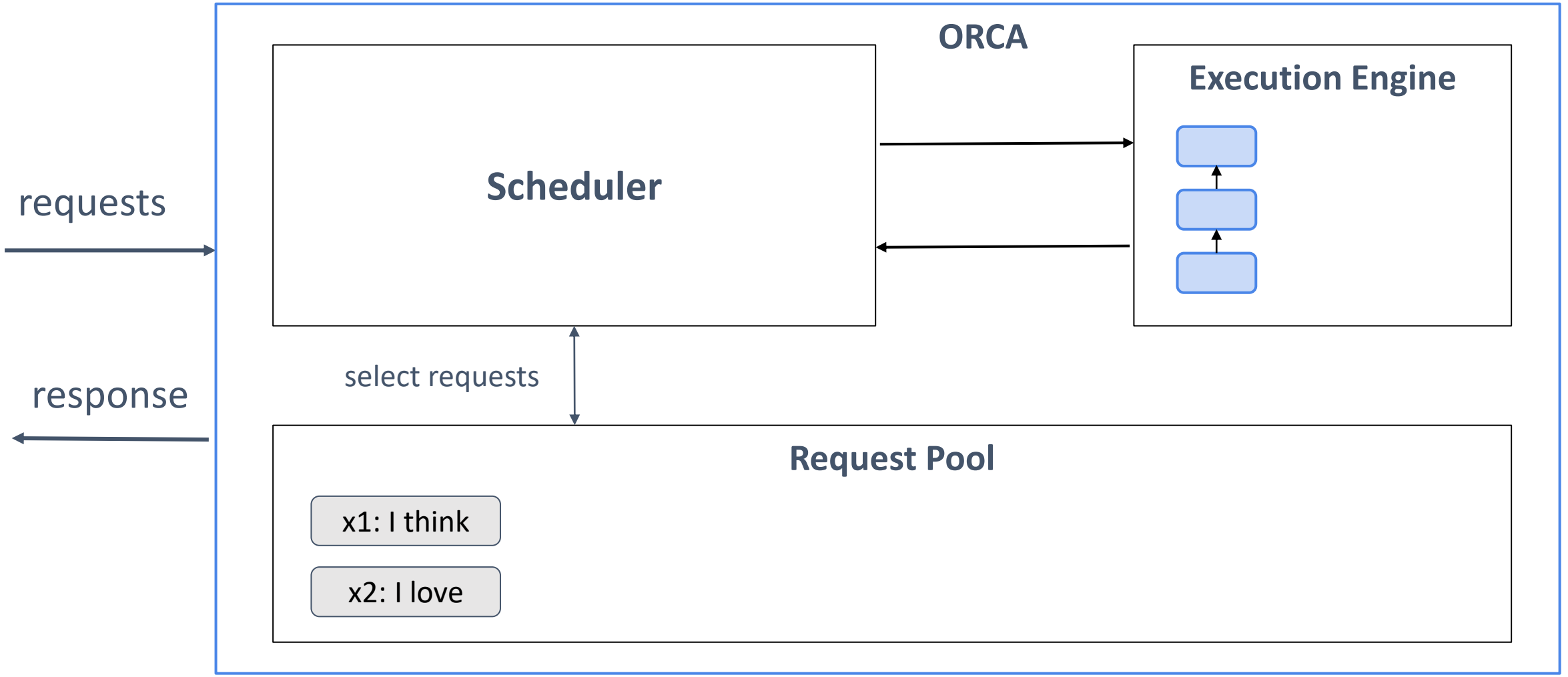

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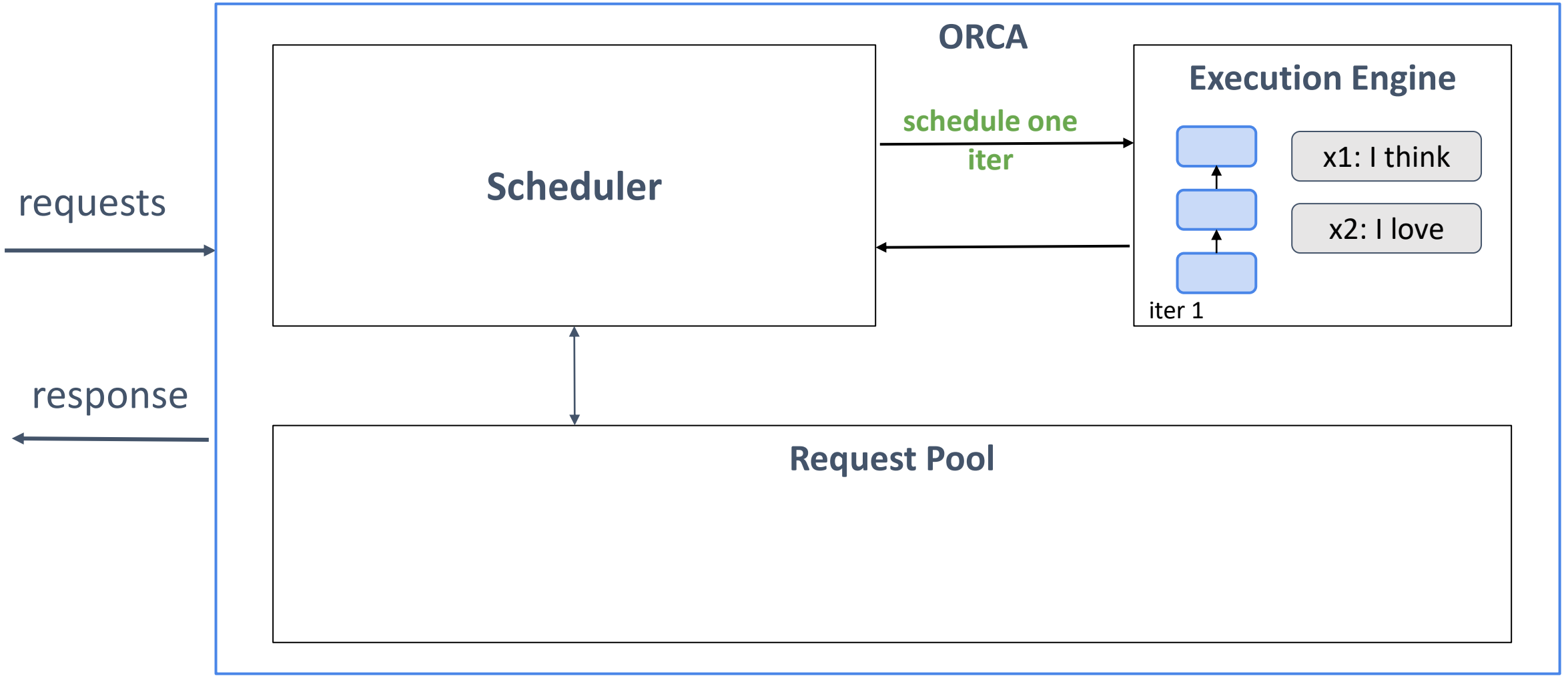

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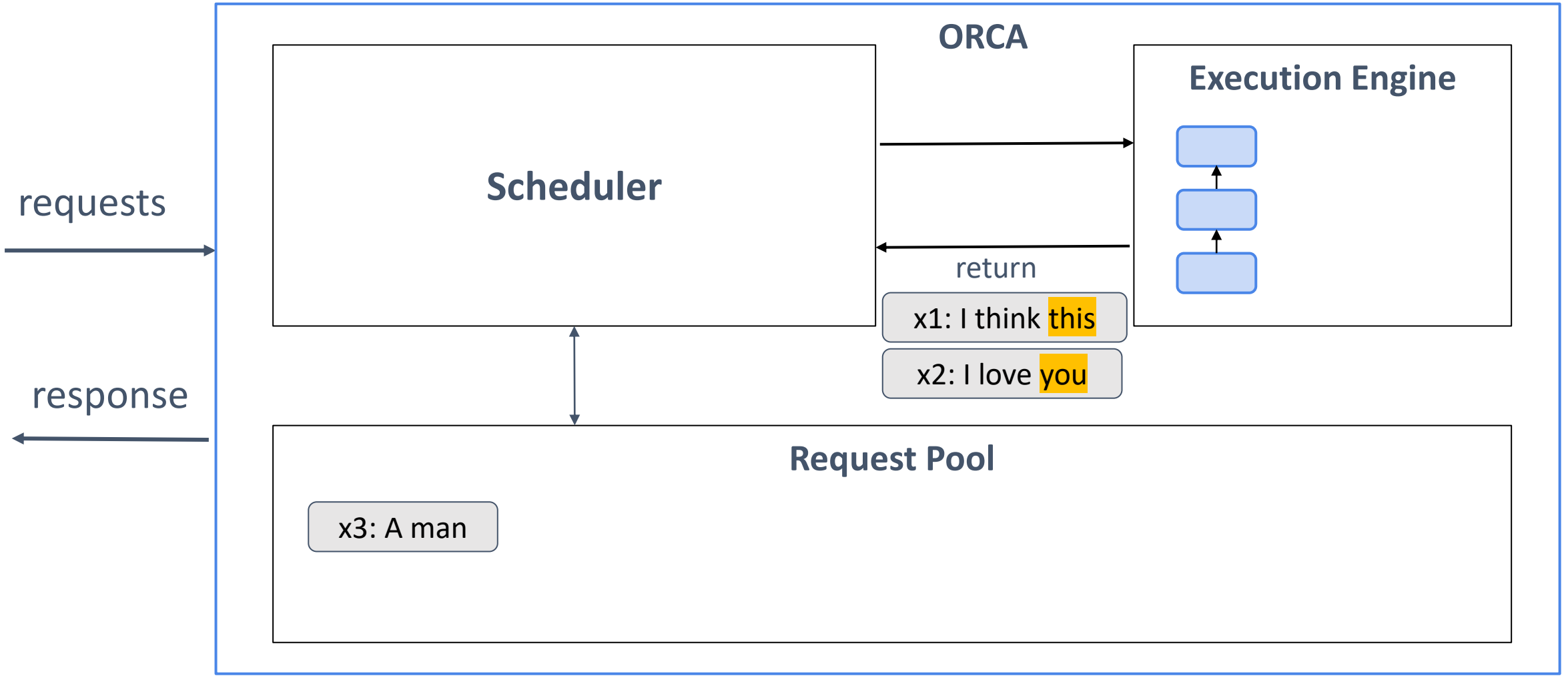

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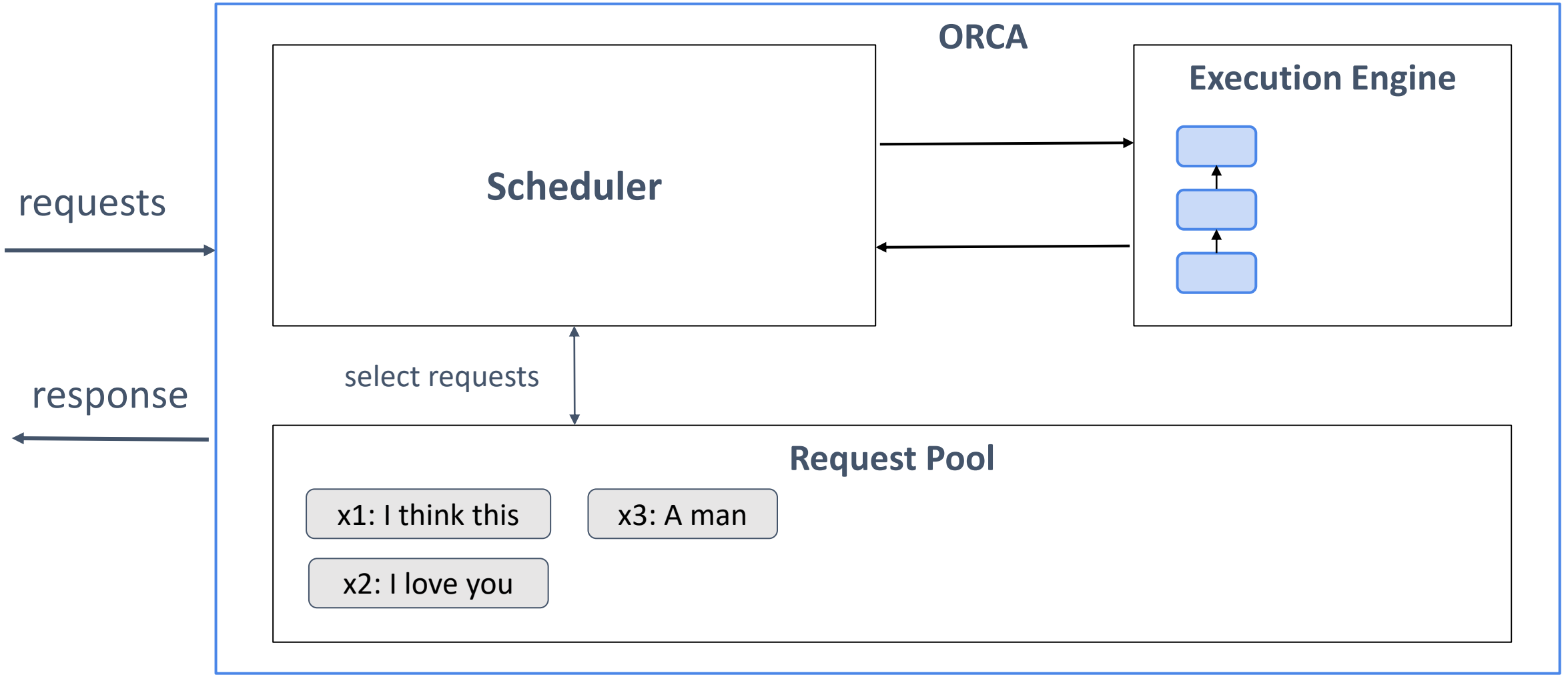

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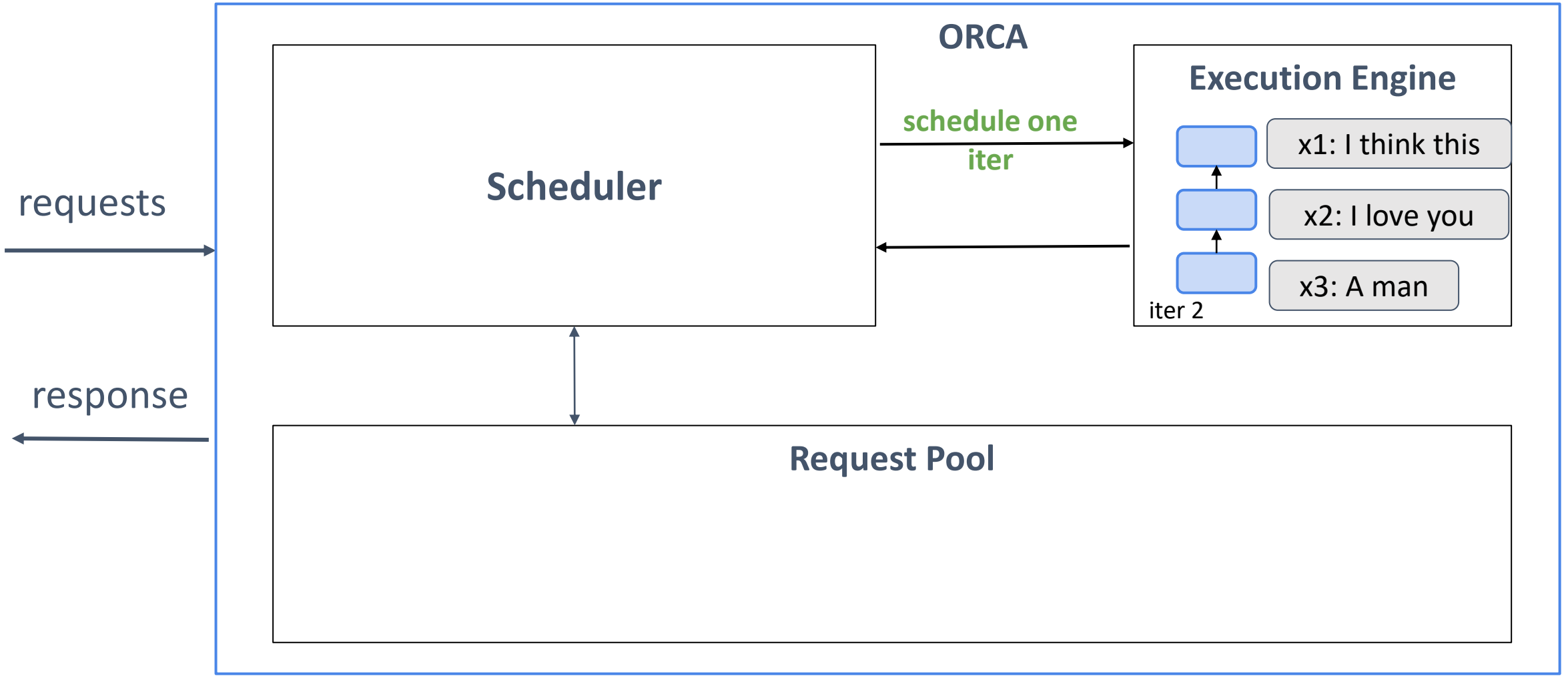

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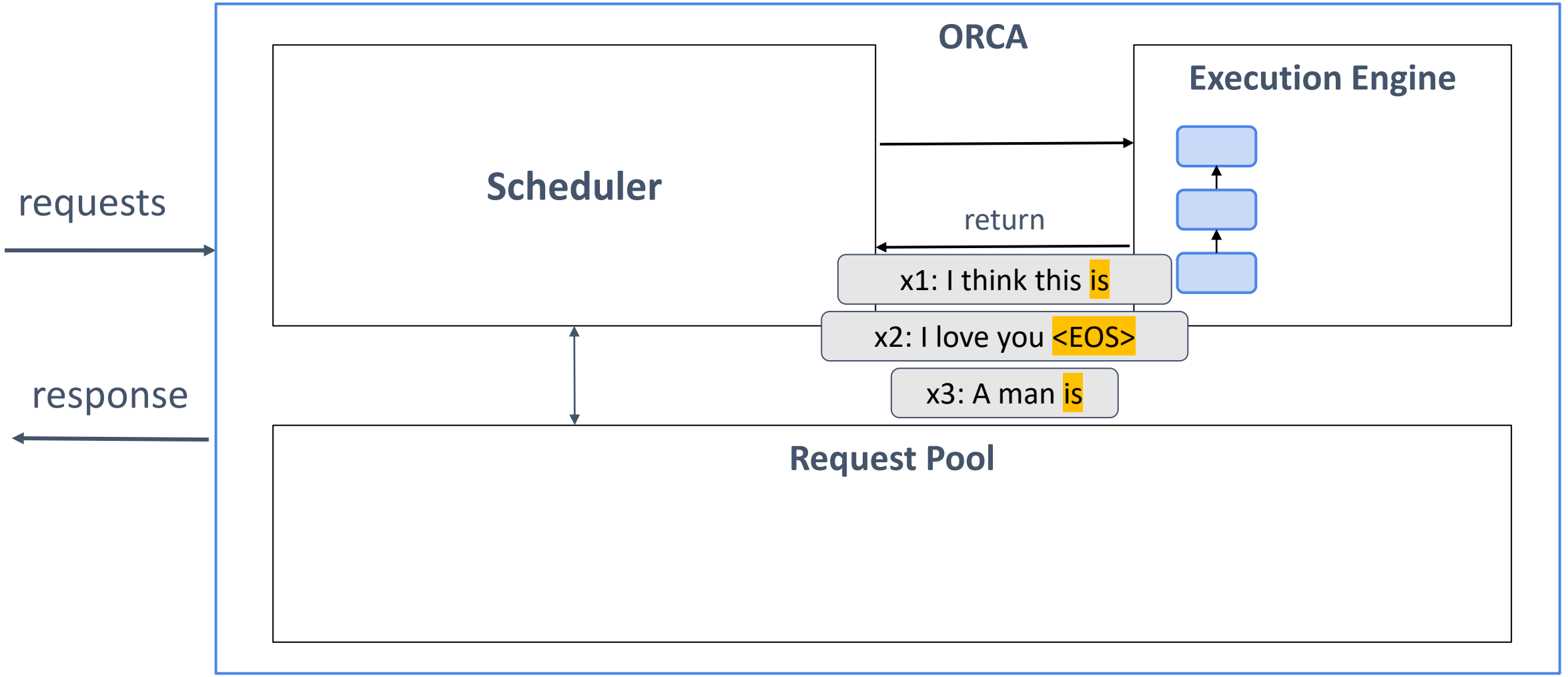

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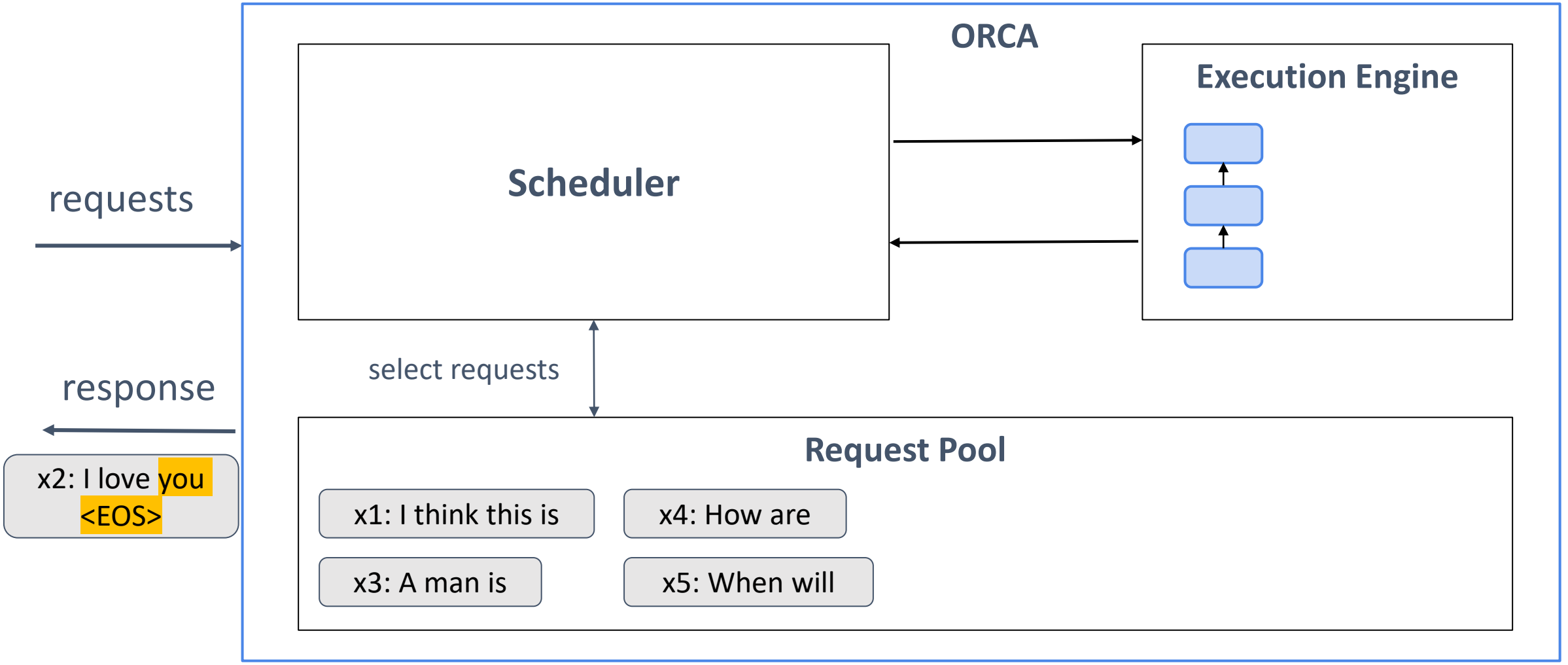

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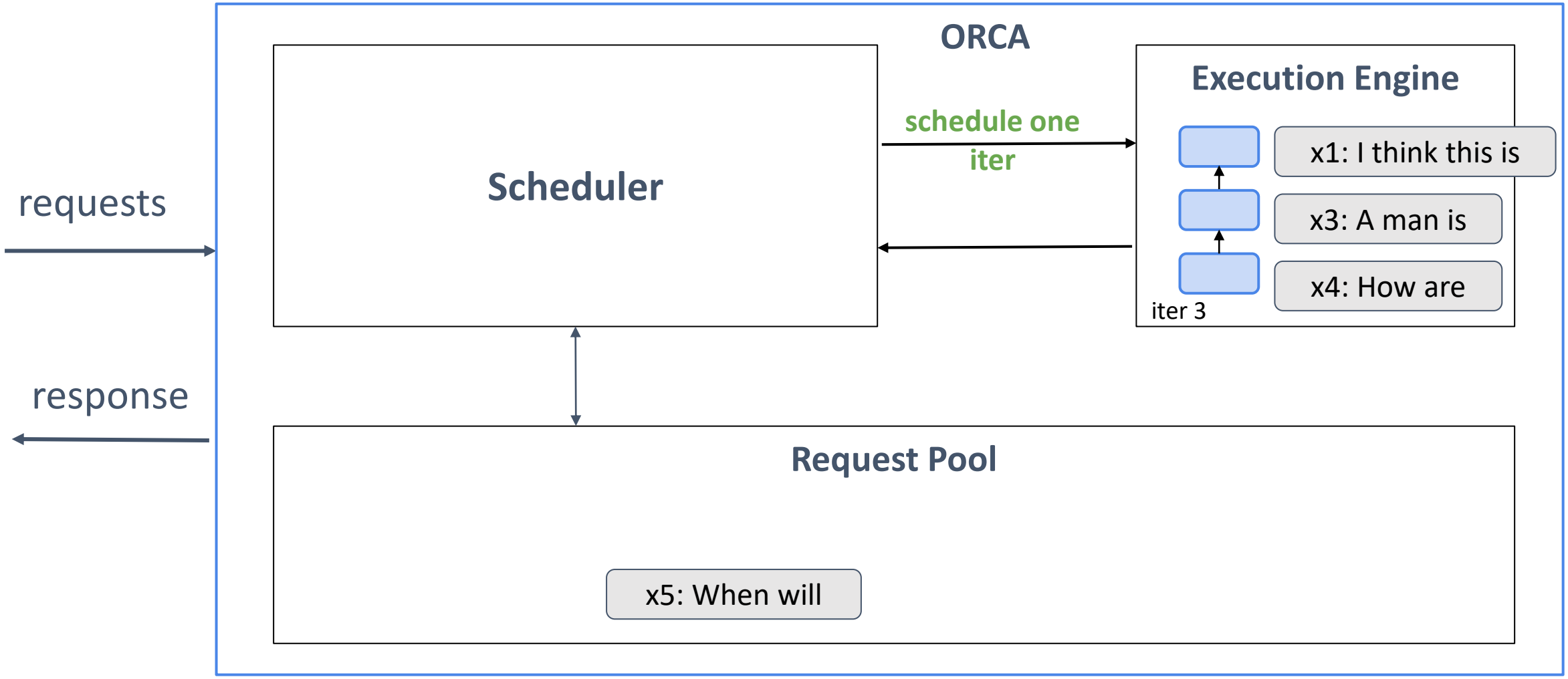

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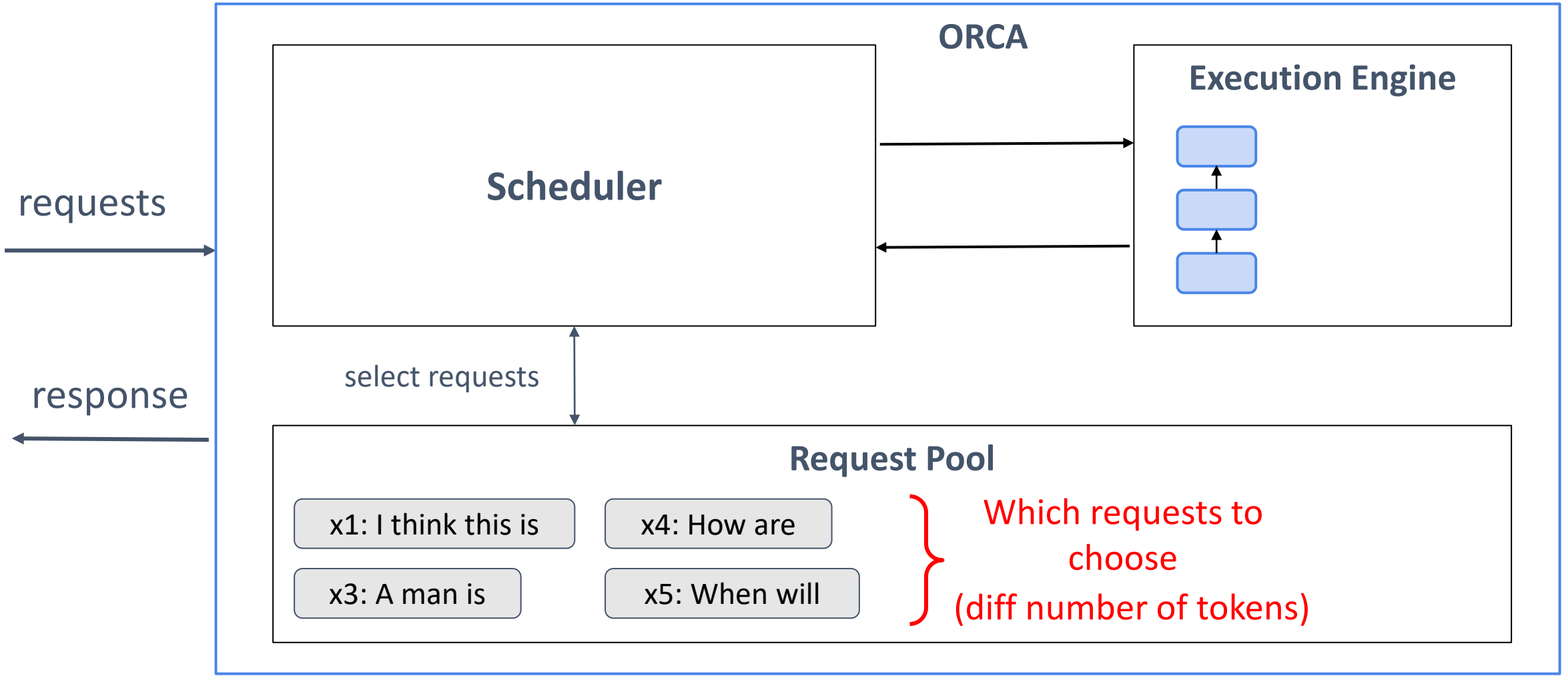


Problem 2: How to Batch Requests?

Let's assume Batch Size B = 1

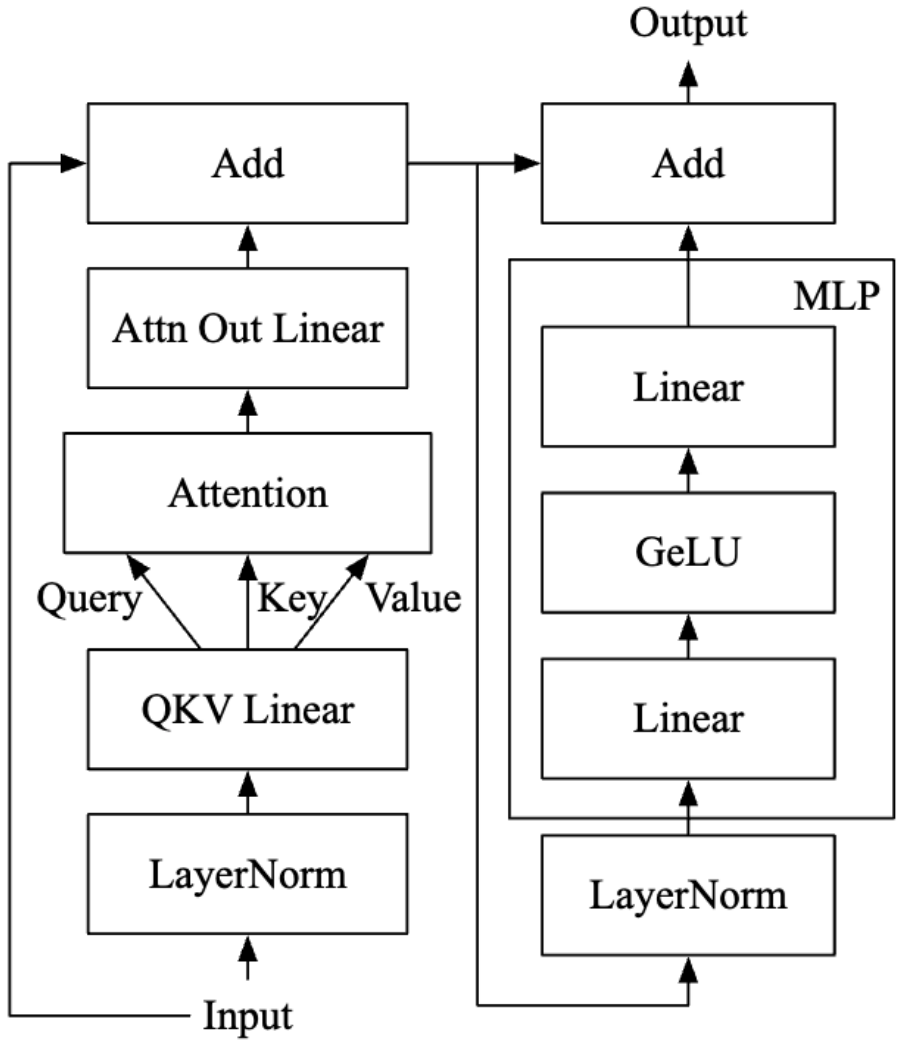
Input Dimension: [L x H] (L=sequence length, H=hidden dim.)

Attention Operation:

1. **QK^T** : [LxH] x [HxL] → [L x L]
2. **P = softmax(QK^T)** : [L x L]
3. **O = PV** : [LxL] x [LxH] → [L x H]

With Batch Size B, **QK^T** will be **[B x L x L]**

With different sequence lengths, QK^T cannot be computed



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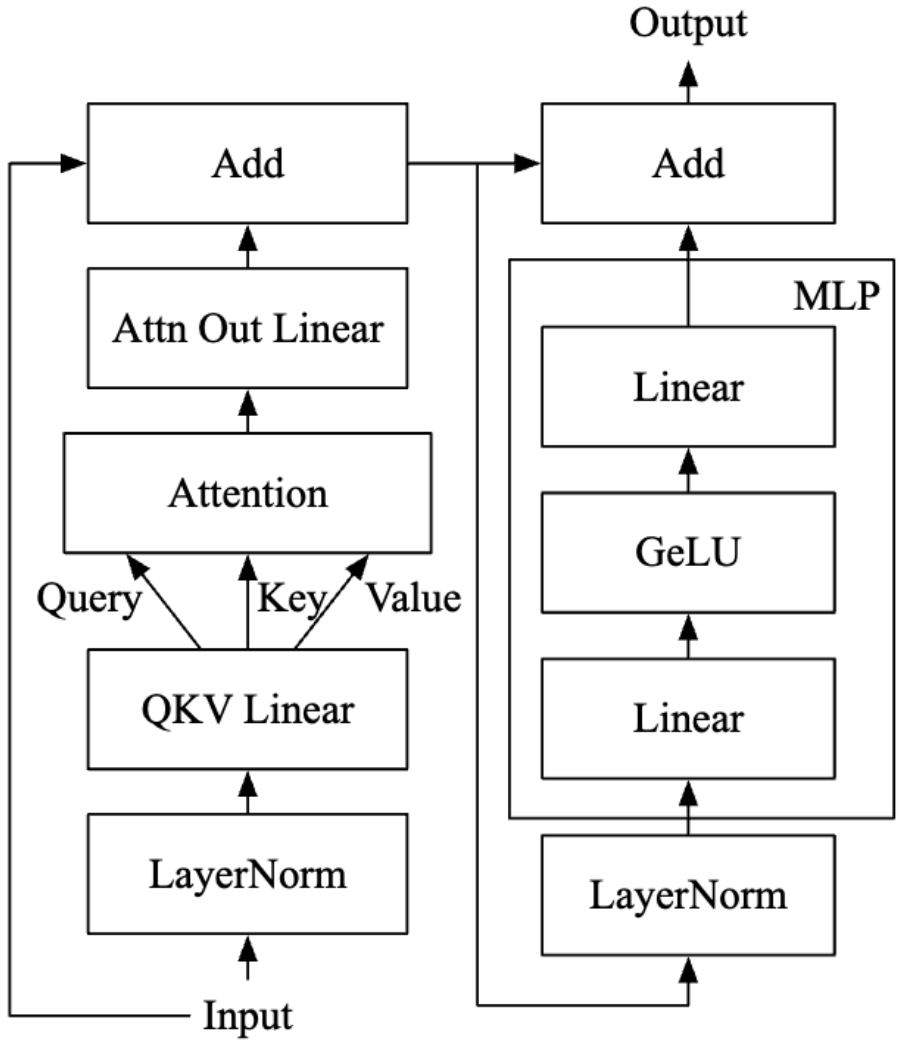


Solution 2: Selective Batching

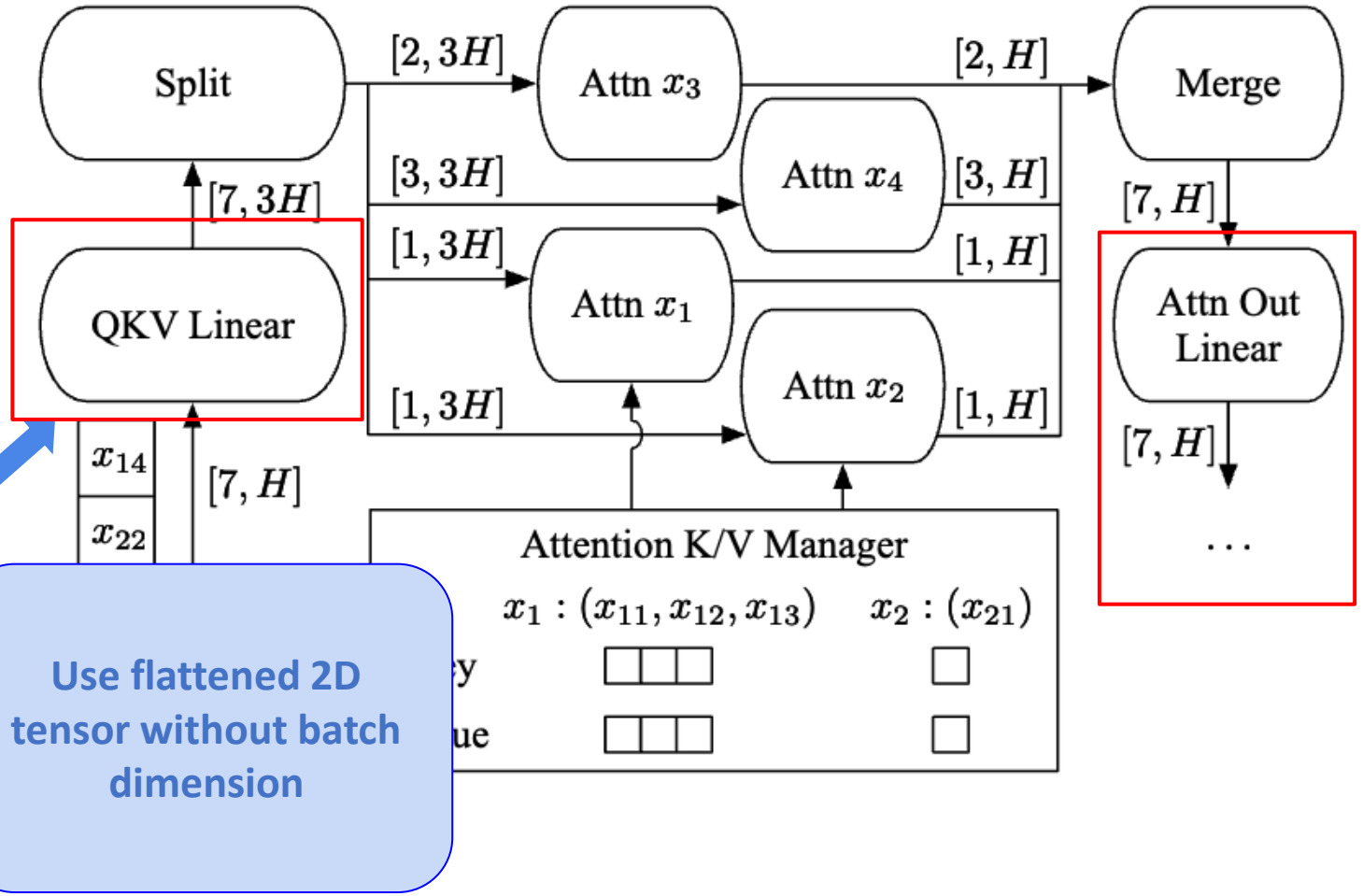
Only **Attention operation** does not work with batching tensors with diff. L_i

Batch for other ops. (Layer Norm, GeLU, etc.)

Coalesce $[L_i, H]$ tensor to $[\Sigma L_i, H]$ for batching

x1: [1,H]
x2: [1,H]
x3: [2,H]
x4: [3,H]

[7,H] tensor



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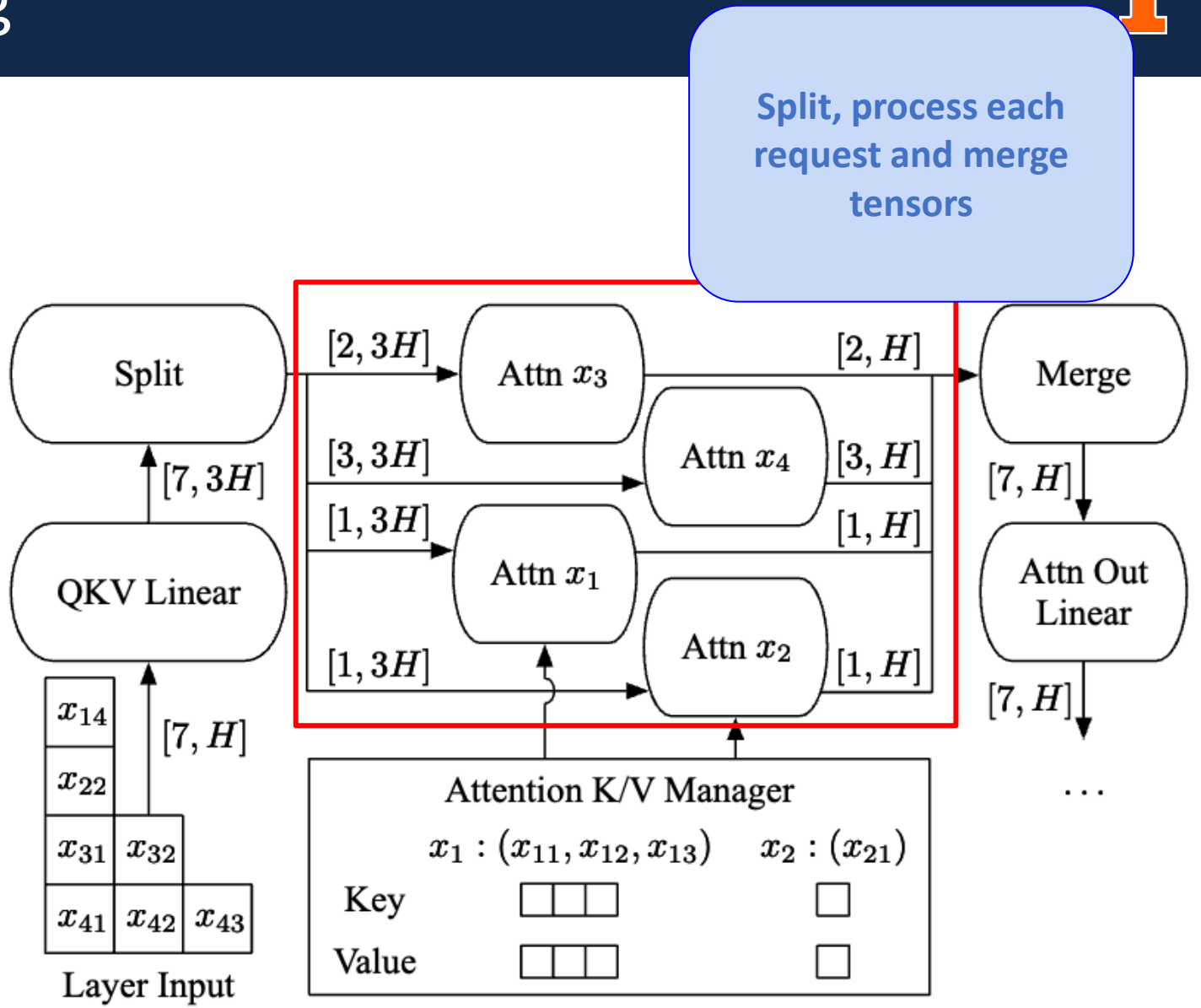
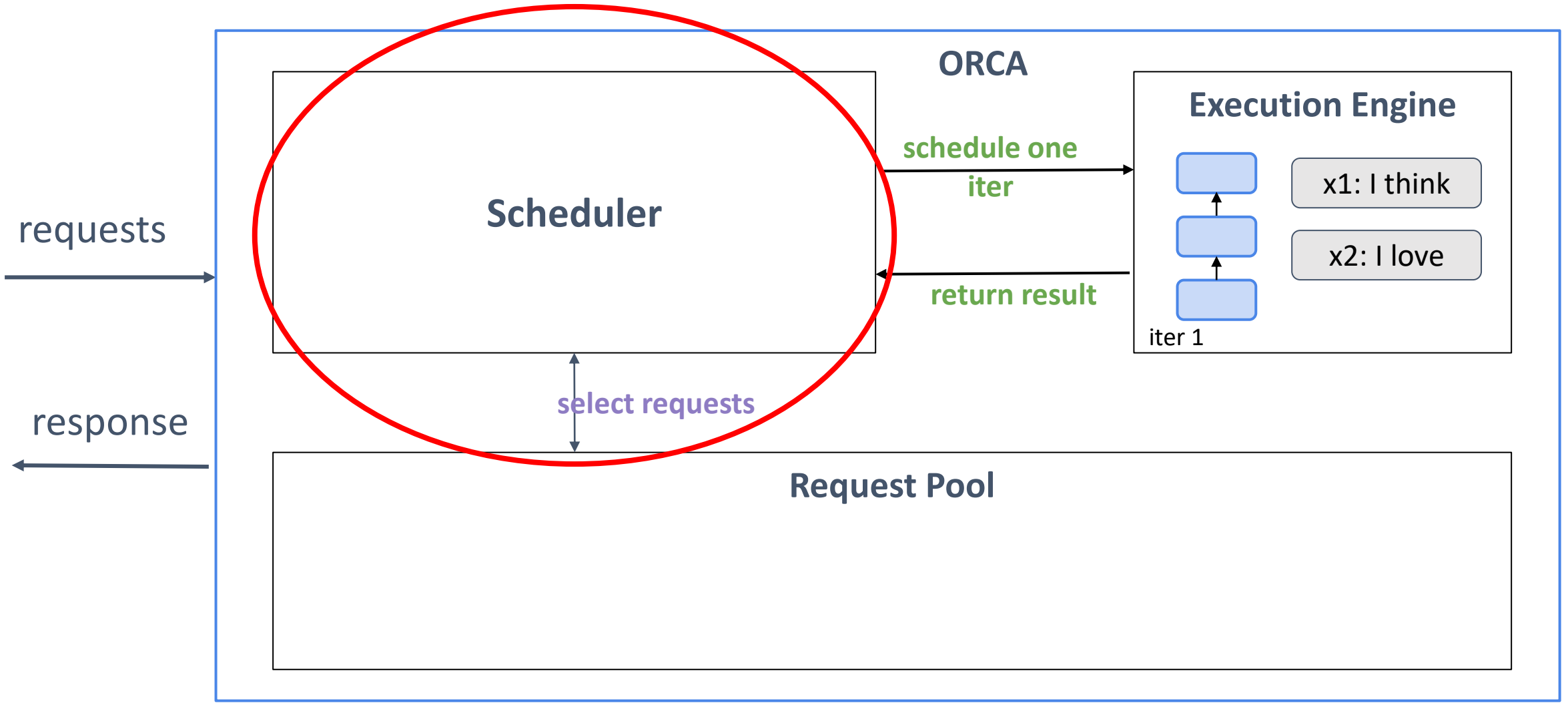


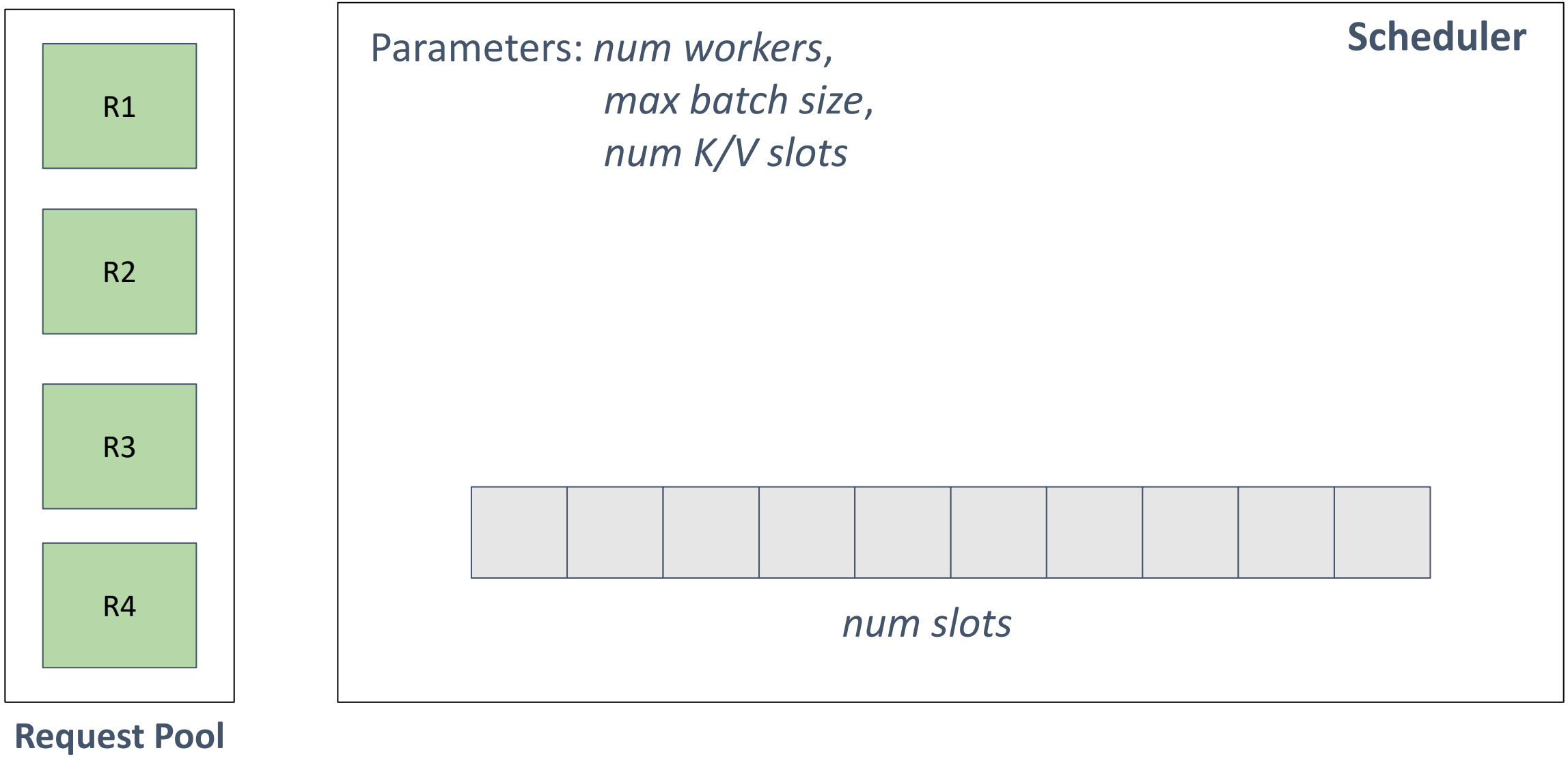
LLM Inference Scheduler



LLM Inference Scheduler

- Enforces iteration-level **first-come-first-served (FCFS)** property
- Maximum batch size → Throughput vs. Latency control knob
- Keep track of number of reserved slots to avoid deadlock
  - Slot := memory required for storing an Attention key and value for a single token
- Reserves max_tokens memory slots per request

Scheduling Algorithm: Selecting a Batch



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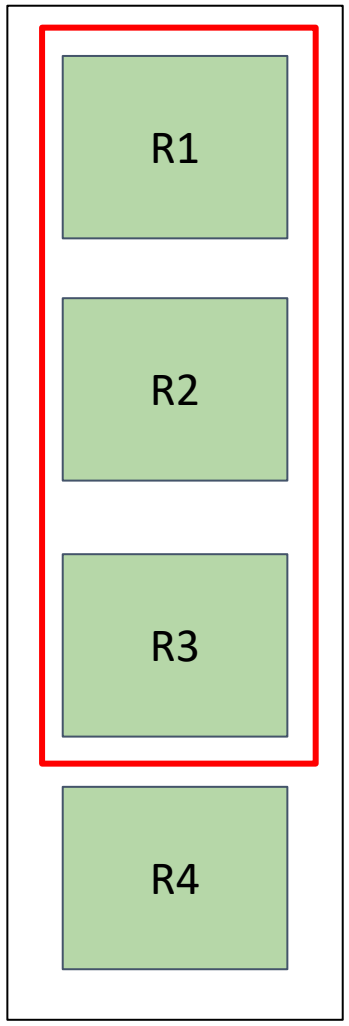


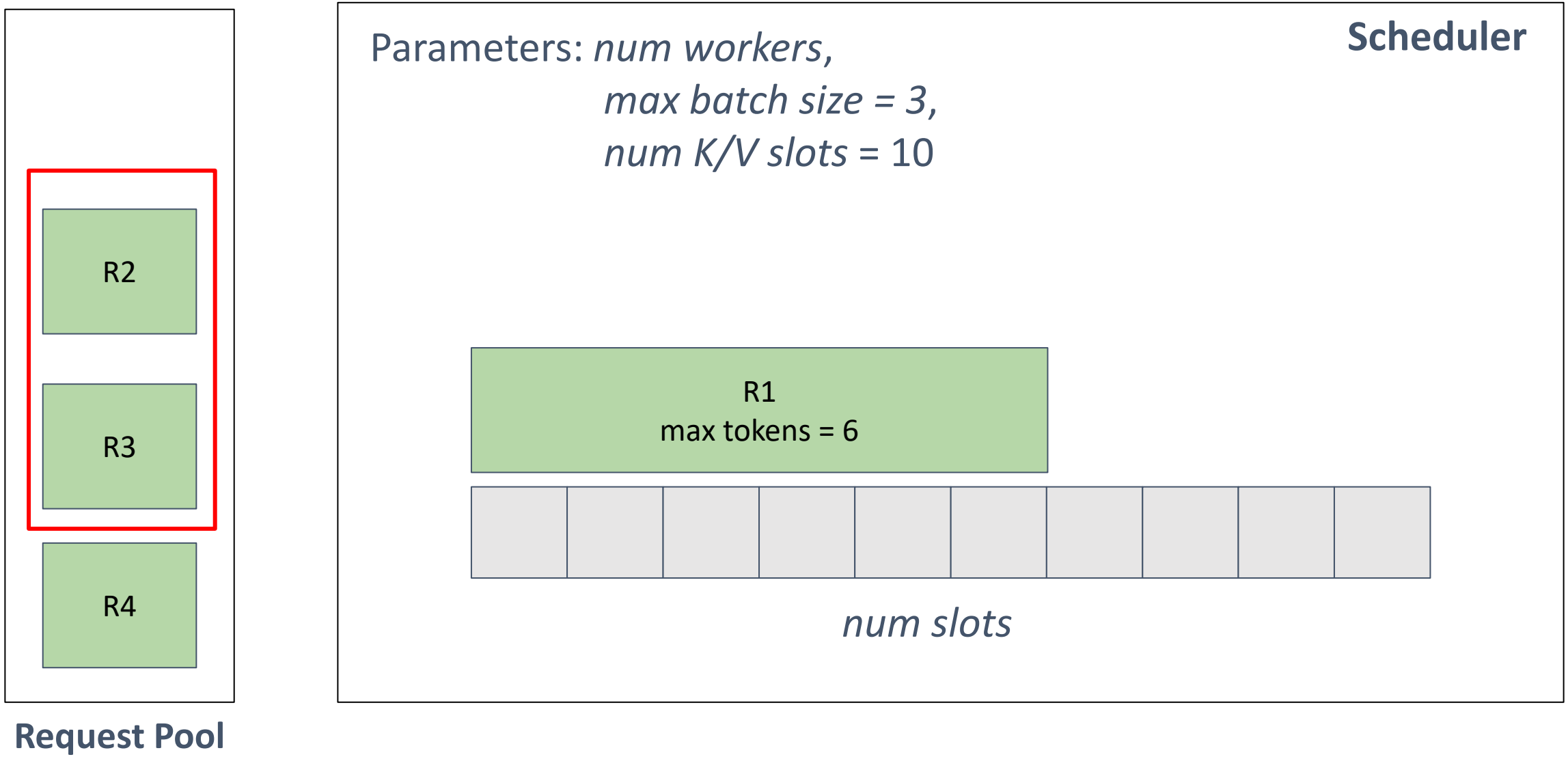
Request Pool

Scheduler

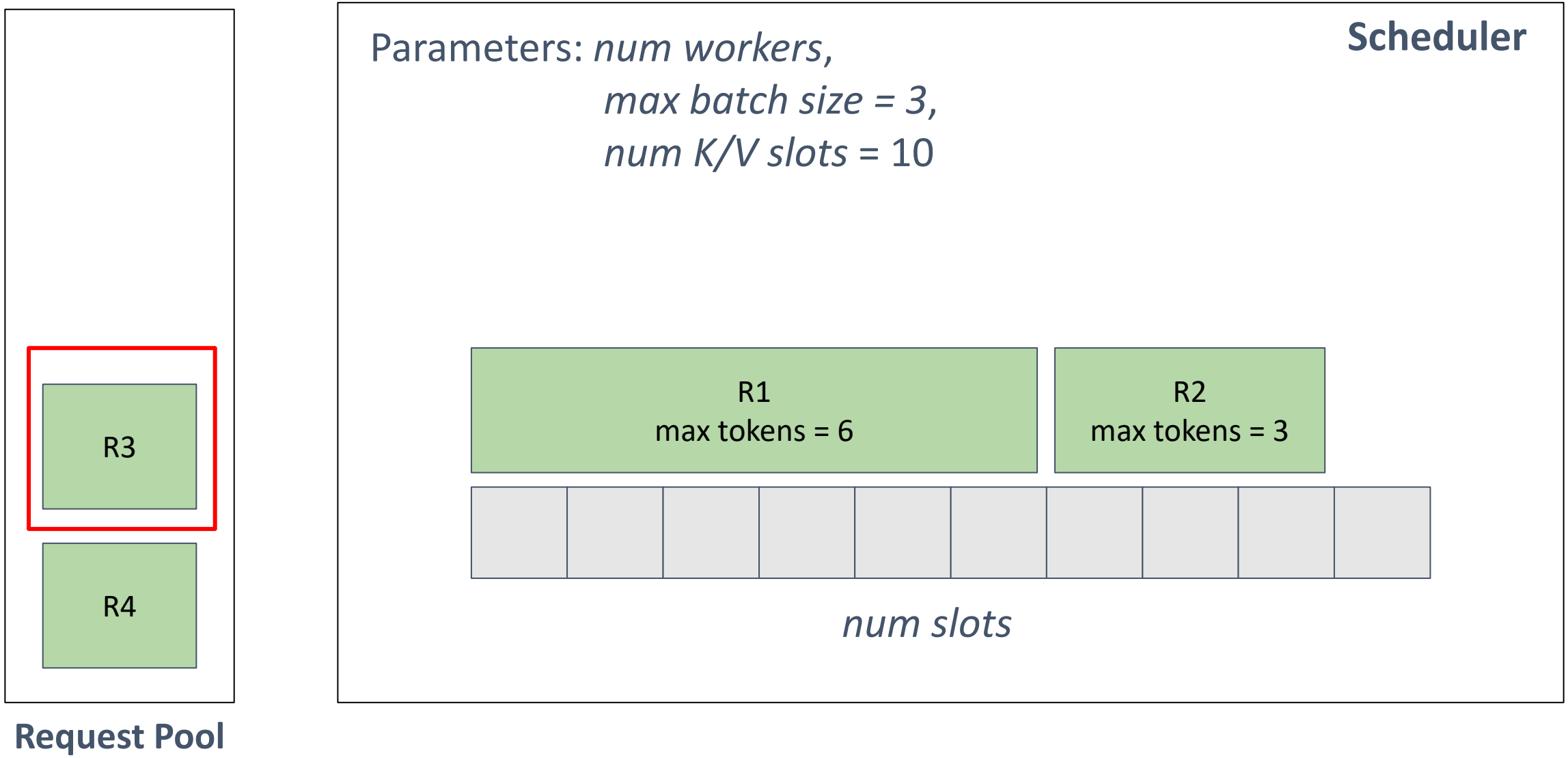
Parameters: *num workers*,
max batch size = 3,
num K/V slots = 10

num slots

Scheduling Algorithm: Selecting a Batch



Scheduling Algorithm: Selecting a Batch



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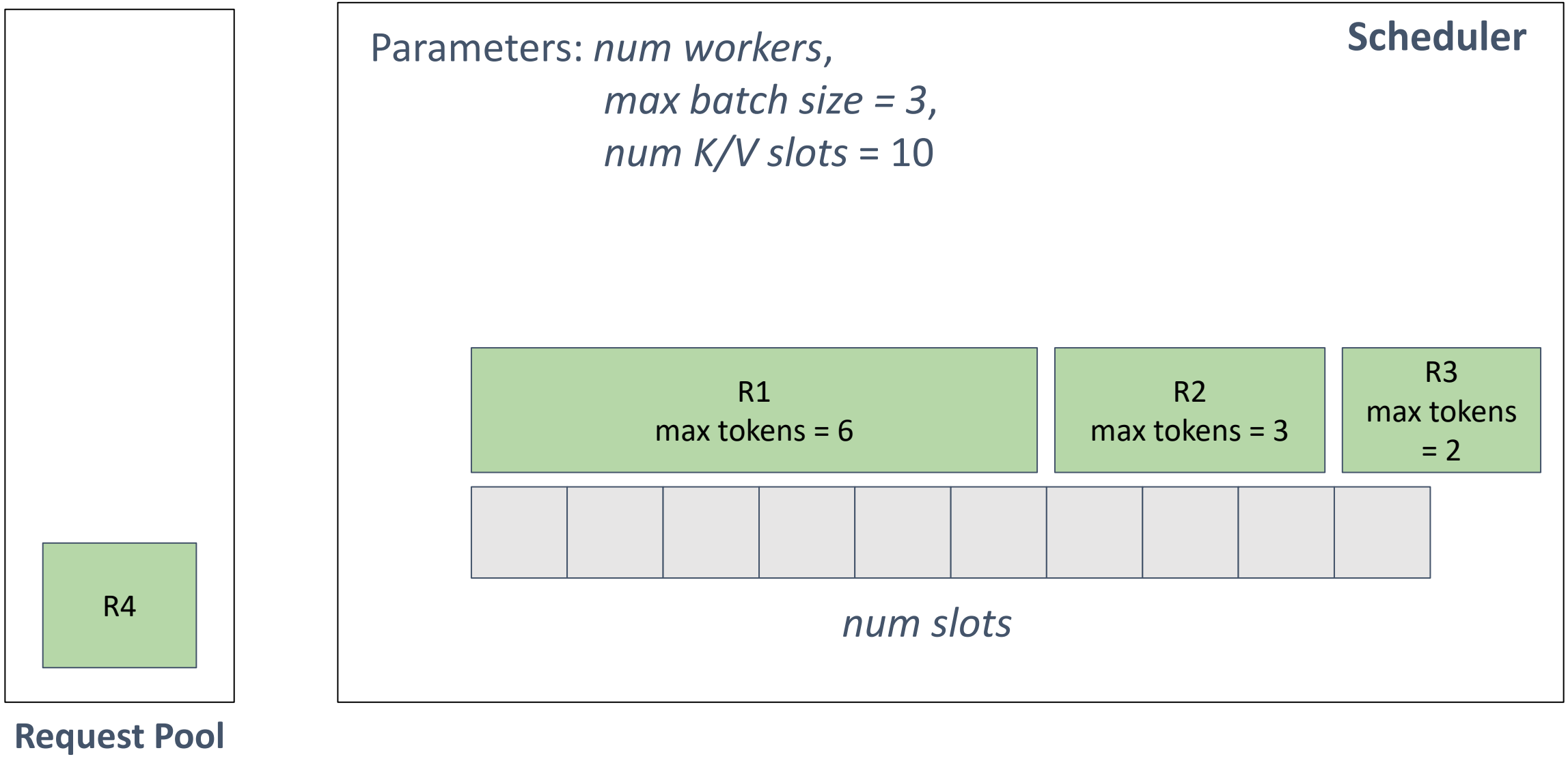

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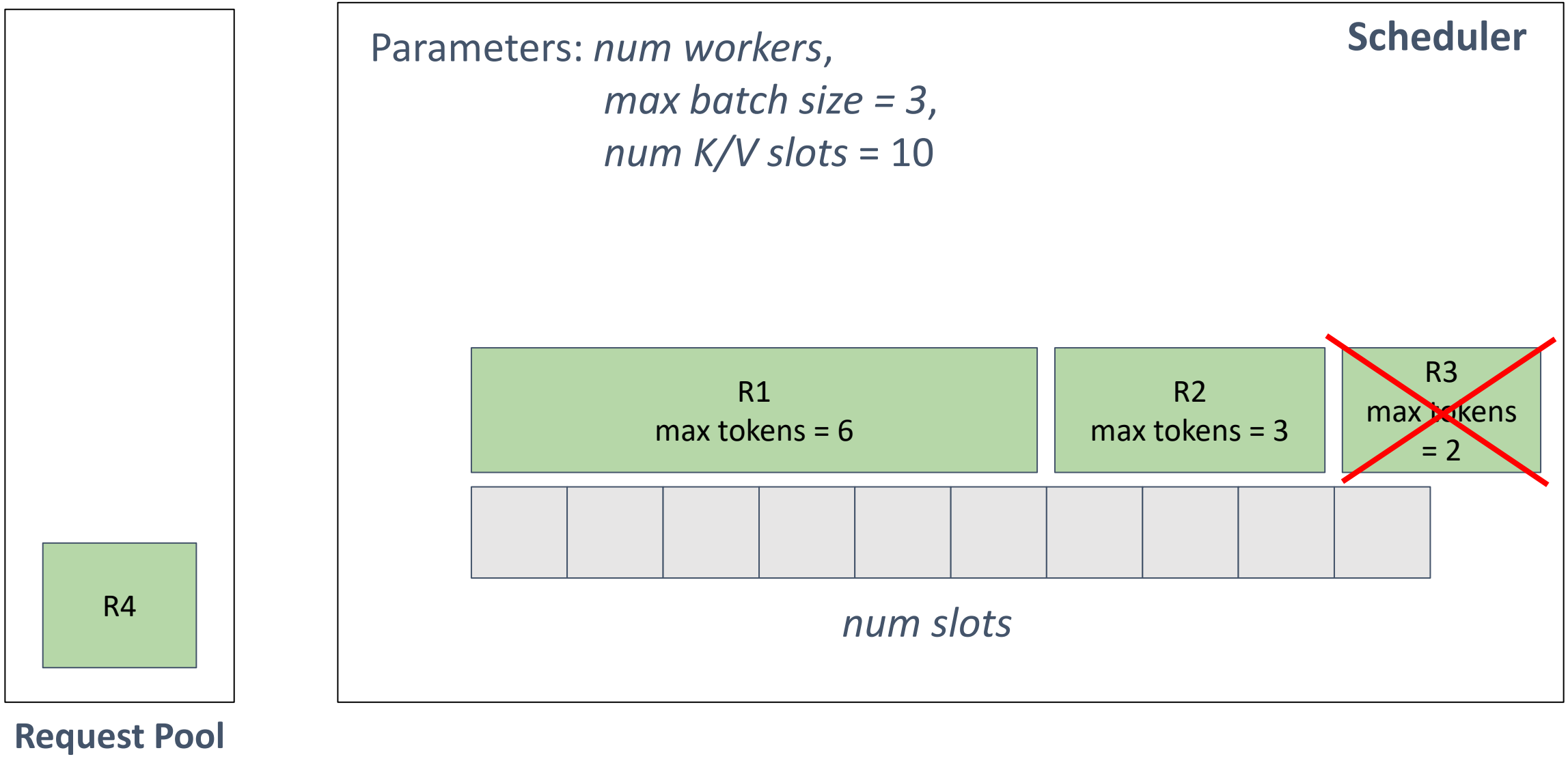

Evaluation Setup

Hardware

4 Azure VMs

Each VM has 8 NVIDIA 40GB A100 GPU connected through NVlink

Models

4 GPT-3 based models (13B, 101B, 175B, 341B)

Inter and Intra-layer Parallelism applied

Baseline

Faster Transformer

Optimized inference engine that supports large scale Transformer models via distributed execution

Evaluation Setup

Hardware

4 Azure V

Each VM has 8
40GB A100
connected th
NVlink

Models

# Params	# Layers	Hidden size	# Inter-partitions	# Intra-partitions
13B	40	5120	1	1
101B	80	10240	1	8
175B	96	12288	2	8
341B	120	15360	4	8

Baseline

Transformer

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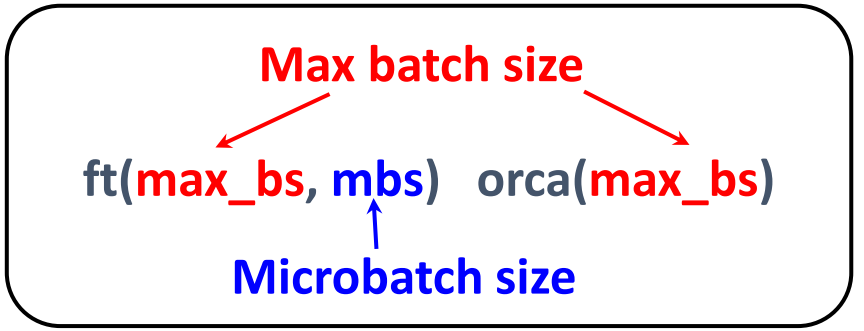
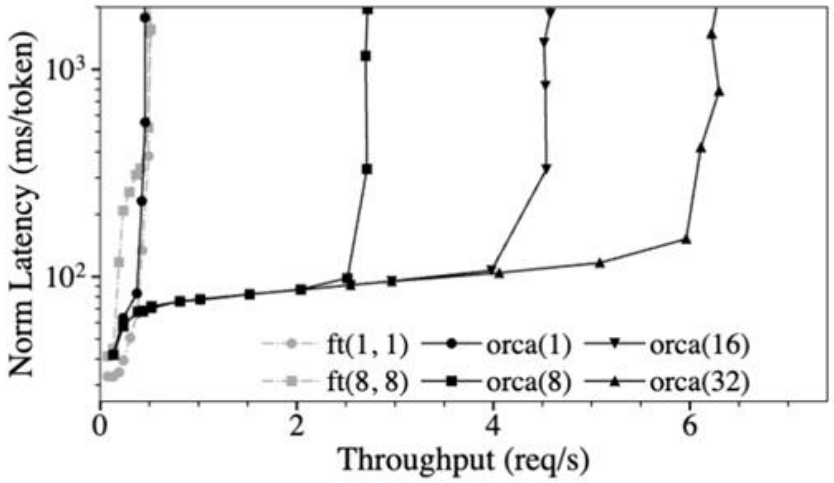
at supports large
Transformer
via distributed
xecution

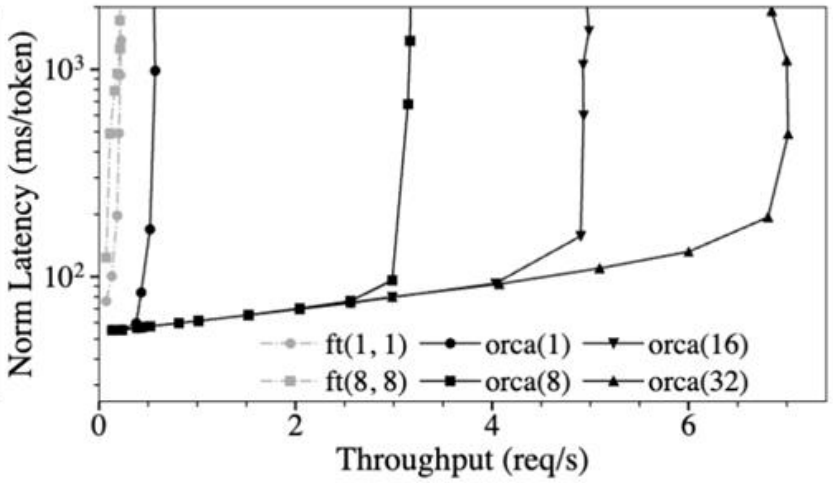
End to End Performance



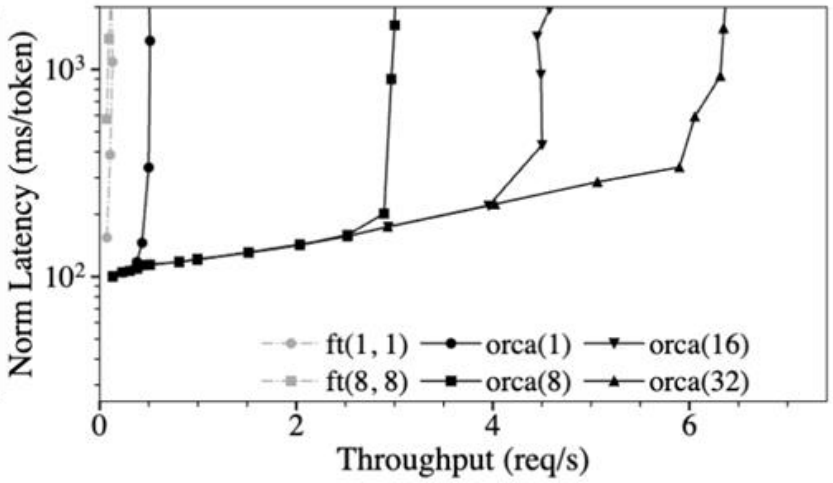
- Request arrival time follows Poisson Distribution
- # Input Tokens ~ U(32, 512)
- # Output Tokens ~ U(1, 128)
- Use custom scheduler used by existing system like Triton for FasterTransformer
- Report median latency normalized by the number of generated tokens of each request



(a) 101B model, 8 GPU.

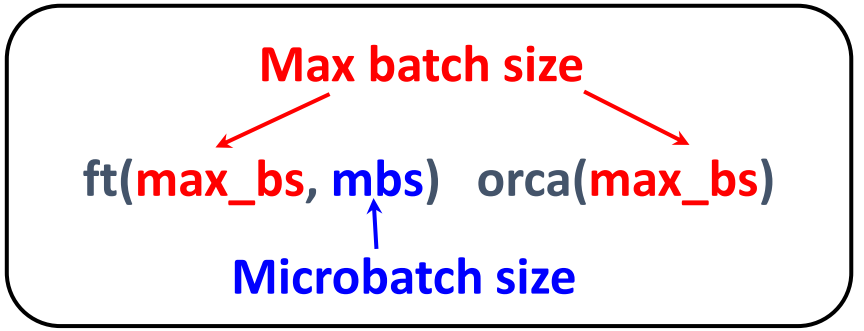


(b) 175B model, 16 GPUs.



(c) 341B model, 32 GPUs.

End to End Performance



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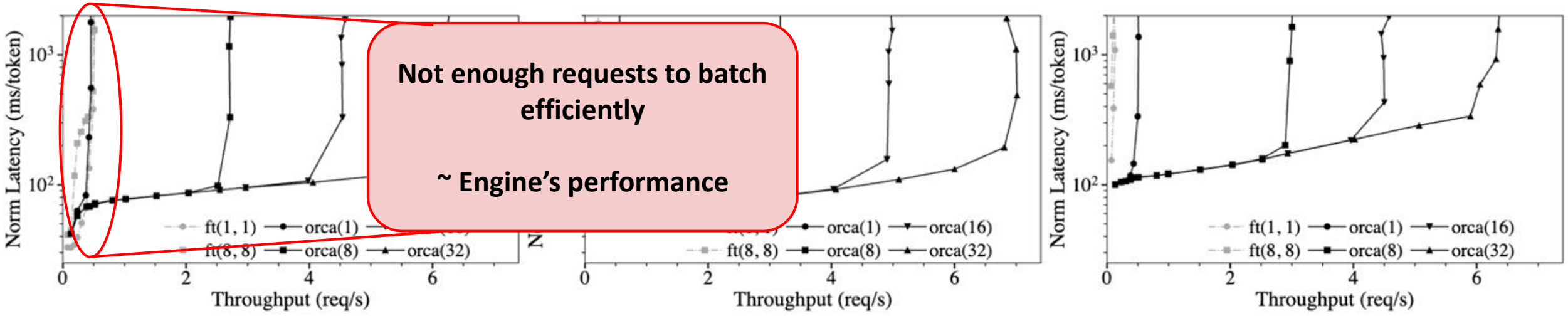

(a) 101B model, 8 GPU. (b) 175B model, 16 GPUs. (c) 341B model, 32 GPUs.

End to End Performance

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- Use custom scheduler used by existing system
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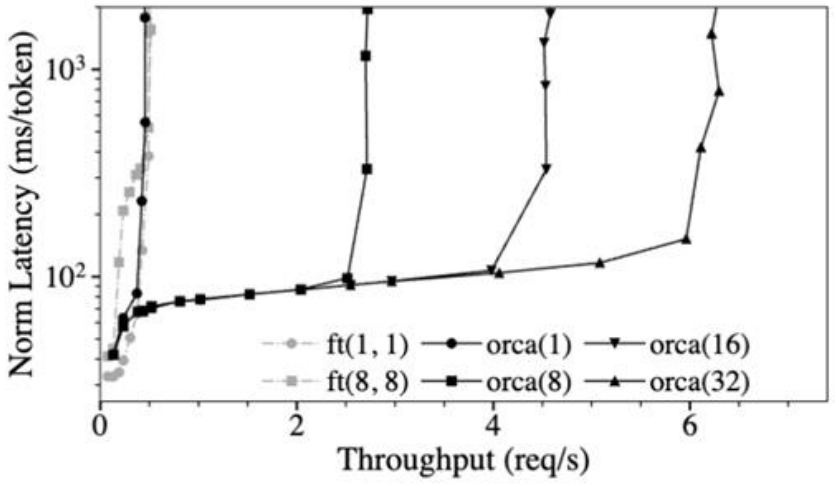
Higher throughput with small increase in latency due to batching



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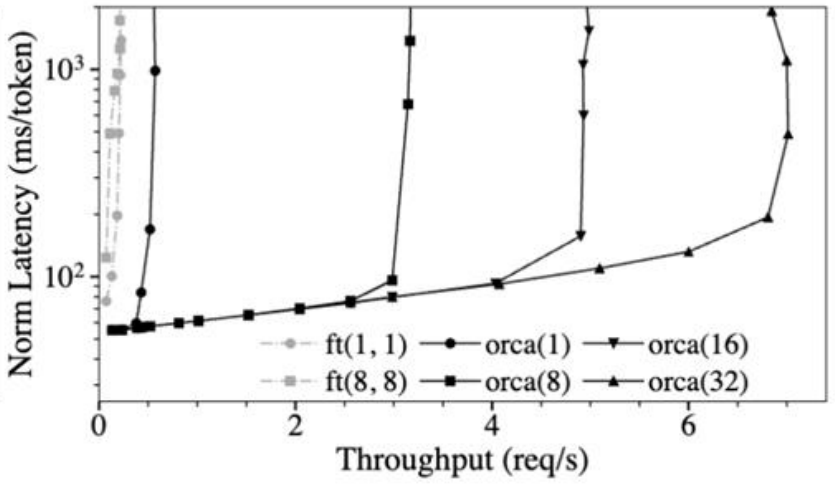
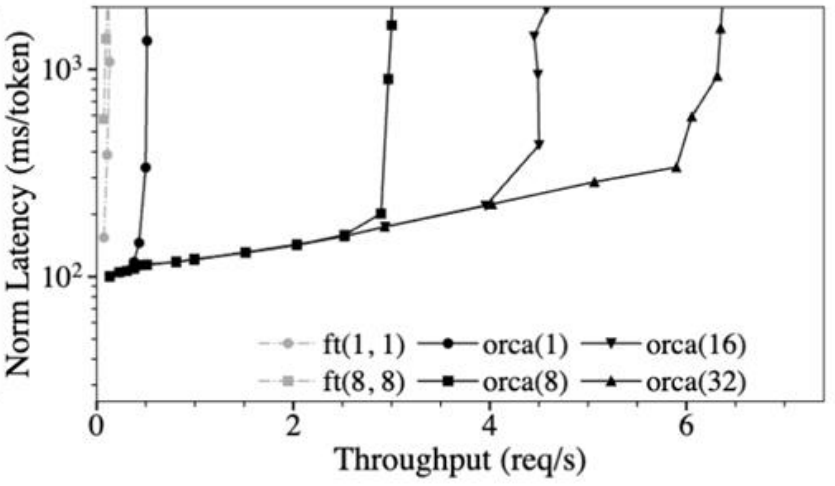
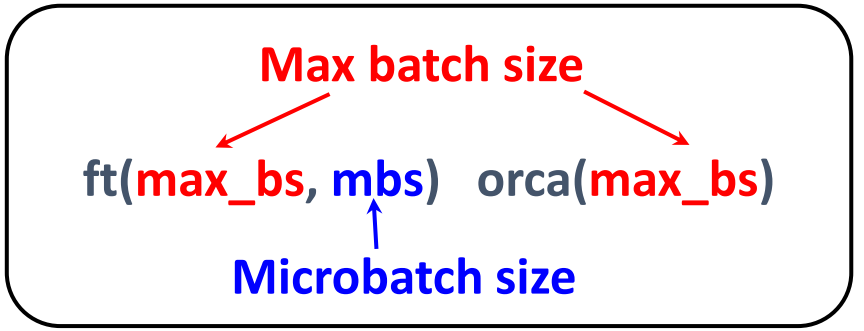


(b) 175B model, 16 GPUs.

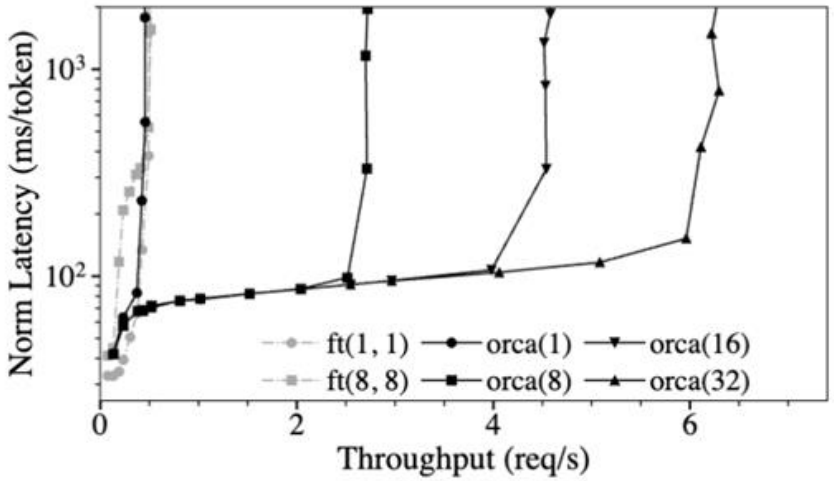


(c) 341B model, 32 GPUs.

Varying Batch Size Configuration



- Increasing batch size improves throughput → early-finished and late-joining requests resolved
- Nevertheless, no guarantee it will not negatively affect latency
- Large batch size → likelihood of different request in batch → FasterTransformer not efficient



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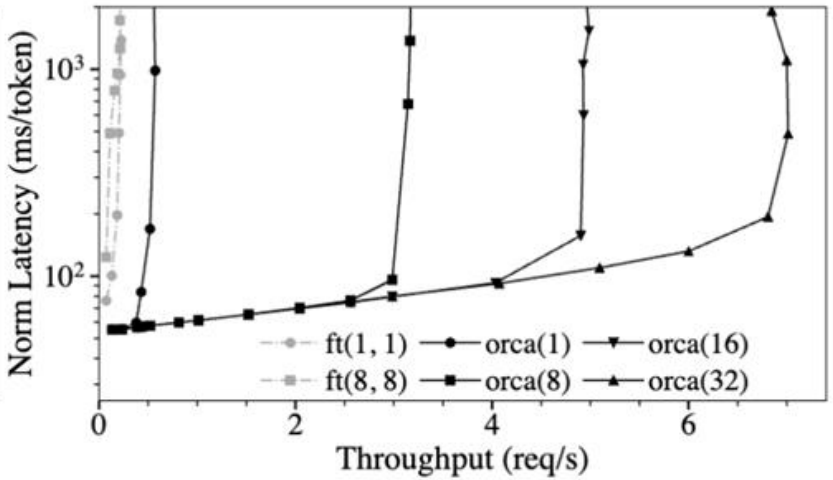

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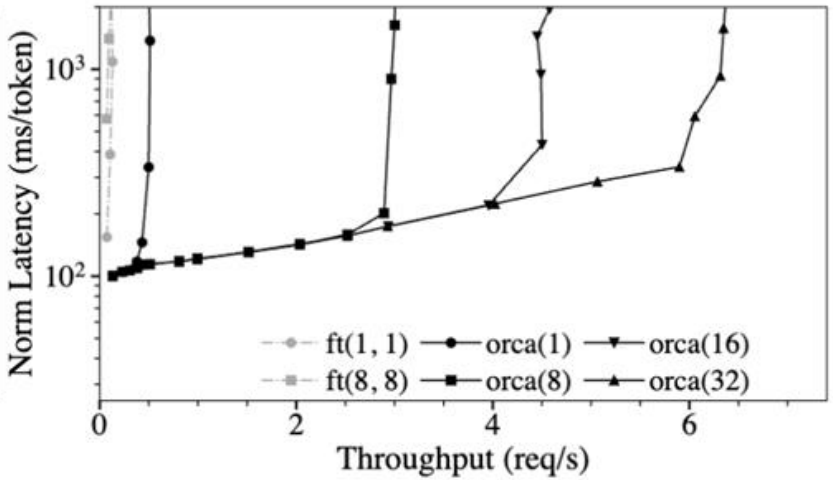

(c) 341B model, 32 GPUs.

Continuous Batching Step-by-Step

- Handle early-finished and late-arrived requests more efficiently
- Higher GPU utilization

Questions?